



Lecture
Notes

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Nuclear Physics

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1 Lecture 01 — Introduction to Nuclear Physics

Overview

Nuclear physics is the branch of physics that studies the atomic nucleus. The nucleus is *not* a point particle: it is a complex, quantum many-body system composed of protons and neutrons (collectively called *nucleons*), bound together by three distinct interactions — the electromagnetic force, the weak interaction, and the nuclear force. Unlike classical many-body systems (e.g. gravitational N -body simulations), there is no scale separation in the nucleus, making exact solutions intractable even on the most powerful computers. Compounding this difficulty, the nuclear force itself is not a fundamental force of nature but rather a residual effect of the strong (QCD) interaction acting between quarks inside nucleons, and no closed-form expression for it exists. This lecture introduces the scope and goals of the course, standard nuclear notation, the Chart of Nuclides, and the characteristic units used in nuclear physics.

Definitions

Physical Concept 1.1: Nucleon

A collective name for the proton and the neutron. A free proton carries electric charge $+e$; a free neutron is electrically neutral. Both have nearly identical masses and are treated in the course as two quantum states of a single object (anticipating isospin symmetry).

Physical Concept 1.2: Atomic Nucleus

A bound quantum many-body system consisting of Z protons and N neutrons (nucleons), with total nucleon number $A = Z + N$ (the *mass number*). Standard notation:

$${}^A_Z\text{X} \quad (1)$$

where X is the chemical symbol of the element. Because the chemical symbol uniquely determines Z , the subscript Z is often omitted. N is almost never written explicitly in the notation, but is given by $N = A - Z$.

Physical Concept 1.3: Isotopes

Two nuclei are *isotopes* of the same element if they share the same proton number Z but have different neutron numbers N . They occupy the same column in the nuclear chart (same Z , different N).

Physical Concept 1.4: Isotones

Two nuclei are *isotones* if they share the same neutron number N but have different proton numbers Z .

Physical Concept 1.5: Isobars

Two nuclei are *isobars* if they share the same mass number $A = Z + N$ while both Z and N individually differ. Isobars lie on anti-diagonal lines in the Chart of Nuclides (constant $Z + N$).

Physical Concept 1.6: Nuclear Force / Nucleon–Nucleon Force

The residual interaction acting between nucleons inside the nucleus. It is *not* a fundamental interaction; it arises as a residue of the colour (QCD) force binding quarks inside each nucleon. Because no first-principles derivation from QCD at nuclear energy scales is currently tractable, the nuclear force is described by *effective models* (Yukawa-type potentials, chiral EFT, etc.). Key empirical properties:

- **Short range:** operates only within $r \lesssim 1$ fm (order of the nucleon radius); can be modelled to zeroth order as a square well potential.
- **Strong:** for $r \lesssim 1$ fm it dominates the Coulomb repulsion between protons.
- **Vanishes outside range:** to good approximation, the nuclear force is zero for $r \gg 1$ fm.

Physical Concept 1.7: Weak Interaction inside the Nucleus

The weak interaction (preferably called the *weak interaction* rather than “weak force”) is responsible for interconverting protons and neutrons:

$$\beta^- \text{-decay: } n \rightarrow p + e^- + \bar{\nu}_e \quad (2)$$

$$\beta^+ \text{-decay: } p \rightarrow n + e^+ + \nu_e \quad (3)$$

The weak interaction acts at a range shorter than the nuclear force itself (it operates *inside* the nucleon).

Physical Concept 1.8: Chart of Nuclides (Nuclear Chart)

A two-dimensional map of all known nuclei, with

- x -axis: neutron number N ,
- y -axis: proton number Z (i.e. the element).

Each cell represents one nuclide ${}^A_Z\text{X}$. Colour coding (one common convention):

- **Black:** stable nuclides (found naturally on Earth).
- **Pink/rose (right of black strip):** β^- emitters — too many neutrons; a neutron converts to a proton, moving one step diagonally toward stability.
- **Blue (left of black strip):** β^+ emitters (or electron capture) — too many protons.
- **Yellow:** α emitters — nucleus ejects a ${}^4\text{He}$ nucleus ($2p + 2n$); the daughter moves two steps down and two steps left in the chart.
- **Green:** fission — nucleus splits into two large fragments.
- **White/empty:** nuclides that either do not exist as bound systems or whose properties have not yet been measured.

The chart can be thought of as a topographic map where “altitude” represents energy; all decay modes correspond to motion toward the black strip (the energy valley of stability).

Physical Concept 1.9: Atomic Mass Unit (u or amu)

The *atomic mass unit* is defined so that the mass of ${}^{12}\text{C}$ is exactly $12 u$:

$$1 u \equiv \frac{1}{12} m({}^{12}\text{C}) \approx 931.5 \text{ MeV}/c^2 \approx 1 \text{ GeV}/c^2. \quad (4)$$

In these units the masses of the free proton and neutron are

$$m_p = 1.00728 u, \quad (5)$$

$$m_n = 1.00867 u. \quad (6)$$

The fractional mass difference $(m_n - m_p)/m_p \approx 0.14\%$ underscores that proton and neutron are nearly identical objects (two isospin states of the nucleon). The electron mass is $m_e \approx 5 \times 10^{-4} u$, roughly $1/2000$ of the nucleon mass, and is usually neglected in nuclear mass budgets.

Key Results

Nuclear size hierarchy. The characteristic length scales are

$$r_{\text{quark}} \sim 0 \text{ (point-like)} \ll r_{\text{nucleon}} \sim 1 \text{ fm} \ll r_{\text{nucleus}} \sim \text{few fm} \ll r_{\text{atom}} \sim 1 \text{ \AA} = 10^{-10} \text{ m.} \quad (7)$$

The nucleus is approximately 1/50,000 the size of the atom.

Nuclear binding energy scale.

$$E_{\text{bind}} \sim \text{few MeV per nucleon} \quad (8)$$

This is 10^6 times larger than the energy scale of chemical bonds ($\sim \text{eV}$), which explains both the enormous energy density of nuclear reactions and the need for the dedicated units below.

Stability line. For light nuclei the stable valley follows approximately $N \approx Z$ (the $N = Z$ line). For heavier nuclei the Coulomb repulsion between protons requires $N > Z$ to maintain stability; beyond $N \approx 50$ the $N = Z$ line lies entirely outside the stable region.

Derivations

No quantitative derivations were performed in this introductory lecture. The lecturer indicated that derivations will begin in Lecture 02 and beyond, after students have completed the assigned reading (Krane, Chapter 3).

Examples

Worked Example 1.1 (Reading nuclear notation)

The nuclide ${}^{56}_{26}\text{Fe}$ (iron-56) has:

- $Z = 26$ protons (the subscript, or equivalently the chemical symbol Fe),
- $A = 56$ nucleons (the superscript),
- $N = A - Z = 30$ neutrons.

Its mass is approximately $56 u$ (to within nuclear binding energy corrections of order $\sim 0.1\%$ per nucleon).

Worked Example 1.2 (Identifying isotopes, isotones, and isobars)

Consider ${}^{12}\text{C}$ and ${}^{13}\text{C}$: both have $Z = 6$, but $N = 6$ and $N = 7$ respectively — they are *isotopes*.

Consider ${}^{12}\text{C}$ ($Z = 6, N = 6$) and ${}^{11}\text{B}$ ($Z = 5, N = 6$): both have $N = 6$ but different Z — they are *isotones*.

Consider ${}^{14}\text{C}$ ($Z = 6, N = 8$) and ${}^{14}\text{N}$ ($Z = 7, N = 7$): both have $A = 14$ but differ in both Z and N — they are *isobars*.

Worked Example 1.3 (The NNDC interactive chart)

The National Nuclear Data Center (NNDC) hosts the interactive Chart of Nuclides at <https://www.nndc.bnl.gov>. Clicking on any nuclide reveals its half-life, decay

modes, excited energy levels, spin-parity J^π , binding energy, and much more. The lecturer emphasised this as the primary numerical reference for the entire course.

Connections

- **Quantum Mechanics II (prerequisite):** The lecturer explicitly stated that students should be familiar with the hydrogen atom, orbital structure, the harmonic oscillator, and finite square-well potentials from QM2. These reappear in the shell model (Lecture ~ 8).
- **Classical mechanics:** N -body gravitational systems (solar system) were cited as an analogy: even three-body classical systems cannot be solved analytically, let alone the quantum many-body nucleus.
- **Cosmological N -body simulations:** Mentioned as the classical many-body benchmark — tractable because of scale separation and a known force (GMM/r^2). The nucleus lacks both advantages.
- **QCD / Standard Model:** The nuclear force is a residue of the QCD colour force. At laboratory nuclear energies QCD cannot be expanded perturbatively (the coupling is $O(1)$), so the nuclear force remains phenomenological. High-energy nuclear experiments (e.g. heavy-ion collisions at the LHC) do probe perturbative QCD.
- **Ab-initio nuclear theory** (Nir Barnea group, HUJI): Mentioned explicitly; performs first-principles calculations starting from nucleon–nucleon potentials, currently limited to light nuclei ($A \lesssim 20\text{--}30$).
- **Nuclear astrophysics:** Stars are the natural environment where nuclear reactions occur at scale; the lecturer identified nuclear astrophysics as one of the major frontiers of contemporary nuclear physics.
- **Solid-state / condensed-matter physics:** The *form factor* concept (topic of Lecture 02 reading) also appears in condensed-matter scattering; students should encounter it there later.

To Remember

1. The nucleus is a **quantum many-body system** — exact solutions are intractable; all nuclear models are effective models.
2. Three interactions act inside the nucleus: **electromagnetic** (repulsive, Coulomb, long-range), **weak** (interconverts p and n , very short-range), and the **nuclear force** (attractive, short-range $\sim 1\text{ fm}$, not derivable from first principles at nuclear energies).
3. Standard notation: ${}^A_Z\text{X}$, with $A = Z + N$. Chemical symbol $\Leftrightarrow Z$.
4. The **unit of length** in nuclear physics is the *femtometre* (fm, also called the “fermi”): $1\text{ fm} = 10^{-15}\text{ m}$.

5. The **unit of mass** is the *atomic mass unit*: $1 u \approx 931.5 \text{ MeV}/c^2$, defined via $m(^{12}\text{C}) \equiv 12 u$.
6. Nuclear **binding energies** are of order $\sim \text{MeV}$ per nucleon — 10^6 times chemical bond energies.
7. Isotopes share Z ; isotones share N ; isobars share A .

Exam / HW Hints

- The exam will be **100% written**, with no homework submission component. However, Dr. Friedman stated that exam problems will be *based on, and may include verbatim, problems from the homework sets* he distributes. Solving all homework problems is therefore the primary exam preparation.
- Past exams from previous lecturers (e.g. Ami Leviatan) are of **limited use** because the course structure is substantially different under Dr. Friedman.
- Some homework problems will involve analysing **real nuclear data** (not only analytical calculations). Familiarity with the NNDC website (nndc.bnl.gov) is expected.
- **Reading before Lecture 02 is mandatory**: Chapter 3 of Krane, *Introductory Nuclear Physics*, with emphasis on **nuclear masses** and the **form factor** (first encounter with the concept).

2 Lecture 02 — Basic Nuclear Properties, Mass Formula, and Liquid Drop Model

Overview

This lecture transitions from the qualitative challenges of modeling the strong nuclear force to empirical methodologies used to extract fundamental nuclear properties. We begin by examining nuclear size and density profiles via electron scattering and form factors. We then exploit the charge-symmetry of the strong force in "mirror nuclei" to isolate Coulomb energy contributions. The lecture formalizes the definition of nuclear masses, binding energy, and nucleon separation energies, followed by a review of modern experimental mass-measurement techniques (e.g., MR-ToF, Penning Traps). Finally, these empirical observations are synthesized into the Semi-Empirical Mass Formula (SEMF) based on the Liquid Drop Model, allowing us to predict nuclear stability and decay pathways along isobaric chains.

1. Nuclear Size, Charge Distribution, and Form Factors

To determine the internal structure and size of a nucleus, we rely on the scattering of high-energy electrons. Electrons are ideal probes because they are fundamental point particles (not subject to the strong force) and interact exclusively via the well-understood electromagnetic and weak interactions. To resolve the nucleus, the electron's de Broglie wavelength λ must be on the order of the nuclear size ($\sim$ fm), requiring beam energies around 100 MeV. At excessive energies, inelastic scattering occurs (creating new particles), which complicates structure probing.

In the framework of scattering theory, the transition matrix element is proportional to the Fourier transform of the potential. The spatial distribution of the nuclear charge $\rho(\vec{r})$ modifies the point-like scattering amplitude by a geometric factor known as the **Form Factor**, $F(\vec{q})$.

Derivation of the Form Factor

Let $\vec{k}$ and $\vec{k}'$ be the initial and final wavevectors of the electron. The momentum transfer is defined as $\vec{q} = \vec{k} - \vec{k}'$. The potential generated by the nuclear charge distribution $\rho(\vec{r})$ dictates the scattering amplitude. Normalizing the charge distribution such that $\int \rho(\vec{r}) d^3r = Ze$, the form factor is defined as:

$$F(\vec{q}) = \frac{1}{Ze} \int \rho(\vec{r}) e^{i\vec{q}\cdot\vec{r}} d^3r$$

The experimentally measured differential cross-section is the point-charge (Mott/Rutherford) cross-section modulated by $|F(\vec{q})|^2$:

$$\left(\frac{d\sigma}{d\Omega} \right)_{\text{exp}} = \left(\frac{d\sigma}{d\Omega} \right)_{\text{point}} |F(\vec{q})|^2 \quad (9)$$

By measuring the cross-section at various angles (and thus various $\vec{q}$), we can invert the Fourier transform to deduce $\rho(\vec{r})$. Experimental results reveal two universal constants for most nuclei: 1. **Nuclear Density** The core density is roughly constant across all nuclei:

$$\boxed{\rho_0 \approx 0.15 \text{ nucleons/fm}^3 \approx 2.5 \times 10^{14} \text{ g/cm}^3} \quad (10)$$

2. ****Nuclear Radius:**** Because the density is constant, the volume scales linearly with the mass number A , meaning the radius scales as:

$$\boxed{R = R_0 A^{1/3}} \quad (11)$$

where $R_0 \approx 1.25$ fm.

2. Mirror Nuclei and the Coulomb Energy Difference

Mirror nuclei are pairs of isobars (same A) where the number of protons and neutrons are swapped, i.e., ${}^A_Z X_N$ and ${}^A_{Z-1} Y_{N+1}$ where $A = 2Z - 1$. Because the strong nuclear force is approximately charge-symmetric (a proton-proton strong interaction is nearly identical to a neutron-neutron strong interaction), the primary difference in binding energy between mirror nuclei arises purely from the electrostatic Coulomb repulsion.

Derivation of Coulomb Energy Difference

Assuming the nucleus is a uniformly charged sphere of radius R , the classical Coulomb self-energy is:

$$E_C(Z) = \frac{3}{5} \frac{(Ze)^2}{4\pi\epsilon_0 R}$$

The energy difference ΔE_C between the mirror nuclei (going from Z to $Z - 1$) is:

$$\begin{aligned} \Delta E_C &= E_C(Z) - E_C(Z - 1) \\ &= \frac{3}{5} \frac{e^2}{4\pi\epsilon_0 R} [Z^2 - (Z - 1)^2] \\ &= \frac{3}{5} \frac{e^2}{4\pi\epsilon_0 R} (2Z - 1) \end{aligned}$$

Since $A = 2Z - 1$ for these specific mirror pairs, and substituting $R = R_0 A^{1/3}$, we get:

$$\Delta E_C = \frac{3}{5} \frac{e^2}{4\pi\epsilon_0 R_0} A^{2/3} \quad (12)$$

By measuring the mass (energy) difference between mirror nuclei, we can extract an independent empirical measurement of R_0 .

3. Masses, Binding Energies, and Separation Energies

In tables, tabulated masses are always ****atomic masses**** (M_{atom}), not bare nuclear masses (M_{nuc}). When calculating energetics, we must account for the electron masses.

$$M_{\text{nuc}}(A, Z) \approx M_{\text{atom}}(A, Z) - Zm_e$$

(Note: We typically neglect the atomic electron binding energies, which are on the order of eV/keV, compared to nuclear energies on the order of MeV).

Nuclear Binding Energy (B)

The Binding Energy is the energy required to completely disassemble a nucleus into its constituent protons and neutrons:

$$\begin{aligned} B(A, Z) &= [Zm_p + Nm_n - M_{\text{nuc}}(A, Z)] c^2 \\ &= [Zm_p + Nm_n - (M_{\text{atom}}(A, Z) - Zm_e)] c^2 \end{aligned}$$

Grouping $m_p + m_e \approx m_H$ (the mass of a hydrogen atom), we write the highly practical formula:

$$B(A, Z) = [Zm_H + Nm_n - M_{\text{atom}}(A, Z)] c^2 \quad (13)$$

Separation Energies and Q -values

The **Q -value** dictates the energy released or absorbed in a reaction. For $X \rightarrow Y + c$, $Q = (M_X - M_Y - M_c)c^2$. Exothermic reactions have $Q > 0$.

The **Neutron Separation Energy** (S_n) is the energy required to remove one neutron (analogous to atomic ionization energy). It is the negative of the Q -value for neutron emission:

$$S_n = [M_{\text{atom}}(A - 1, Z) + m_n - M_{\text{atom}}(A, Z)] c^2$$

Similarly, the **Proton Separation Energy** (S_p) is:

$$S_p = [M_{\text{atom}}(A - 1, Z - 1) + m_H - M_{\text{atom}}(A, Z)] c^2$$

4. Experimental Mass Measurement

Given that typical binding energies per nucleon are ~ 8 MeV, and nuclear masses are on the scale of GeV, precision mass measurement is critical ($\sim$ keV accuracy).

- **Standard Mass Spectrometer:** Ions enter a Wien filter (crossed $\vec{E}$ and $\vec{B}$ fields). Only particles with velocity $v = \mathcal{E}/B$ pass un-deflected. They then enter a uniform magnetic field $\vec{B}'$, executing uniform circular motion where the Lorentz force provides the centripetal acceleration:

$$qvB' = \frac{mv^2}{R} \implies R = \frac{mv}{qB'} \implies \frac{m}{q} = \frac{RBB'}{\mathcal{E}}$$

By measuring the radius R , the mass-to-charge ratio is determined.

- **MR-ToF (Multi-Reflection Time-of-Flight):** Ions are trapped in an electrostatic potential well, bouncing back and forth. Because kinetic energy is uniform, velocity varies inversely with $\sqrt{m}$. Over many reflections, the flight time differences amplify: $t \propto \sqrt{m}$. This yields ultra-fast ($\sim$ ms), high-precision separation, ideal for short-lived exotic nuclei.
- **Penning Traps:** Ions are confined by a combination of a strong axial magnetic field and a quadrupole electric field. The precise measurement of the cyclotron frequency $\omega_c = qB/m$ via phase shifts allows for mass measurement accuracies down to $\Delta m/m \sim 10^{-10}$.

5. The Semi-Empirical Mass Formula (SEMF)

To model the systematic trends in the nuclear binding energy (e.g., the saturation of $B/A \approx 8$ MeV), we employ the **Liquid Drop Model**, treating the nucleus as an incompressible fluid droplet with several distinct physical contributions.

$$B(A, Z) = a_v A - a_s A^{2/3} - a_c \frac{Z(Z-1)}{A^{1/3}} - a_a \frac{(A-2Z)^2}{A} + \delta(A, Z) \quad (14)$$

Physical meaning of the terms:

1. **Volume Term** ($+a_v A$): Represents the strong nuclear force. Since B/A is roughly constant, each nucleon only interacts with its nearest neighbors, meaning the binding energy scales linearly with the volume (and thus A).
2. **Surface Term** ($-a_s A^{2/3}$): Nucleons on the surface have fewer neighbors than those in the bulk. This reduces the total binding energy proportional to the surface area $\propto R^2 \propto A^{2/3}$.
3. **Coulomb Term** ($-a_c Z(Z-1)A^{-1/3}$): Electrostatic repulsion between the Z protons. Scales as $1/R \propto A^{-1/3}$.
4. **Asymmetry Term** ($-a_a (A-2Z)^2/A$): Quantum mechanical in origin (Pauli Exclusion Principle). Nuclei favor $N = Z$. Any deviation costs energy.
5. **Pairing Term** (δ): Empirical observation that nucleons favor pairing up (spin up with spin down).

$$\delta(A, Z) = \begin{cases} +a_p A^{-1/2} & \text{even } Z, \text{ even } N \text{ (most stable)} \\ 0 & \text{odd } A \\ -a_p A^{-1/2} & \text{odd } Z, \text{ odd } N \text{ (least stable)} \end{cases}$$

Example Calculation for $^{56}_{26}\text{Fe}$: Using the typical coefficients: $a_v = 15.5$, $a_s = 16.8$, $a_c = 0.72$, $a_a = 23$, $a_p = 12$ (all in MeV). For ^{56}Fe , $Z = 26$, $N = 30$ (even-even $\Rightarrow$ positive pairing).

$$\begin{aligned} B(56, 26) &= 15.5(56) - 16.8(56)^{2/3} - 0.72 \frac{26(25)}{56^{1/3}} - 23 \frac{(56-52)^2}{56} + 12(56)^{-1/2} \\ &\approx 496 \text{ MeV} \end{aligned}$$

Thus, $B/A \approx \frac{496}{56} \approx 8.9$ MeV, residing perfectly at the peak of the binding energy curve.

Isobaric Chains and Parabolas

If we hold A constant (an isobaric chain) and plot Mass vs. Z , the SEMF reveals a quadratic relationship: $M(A, Z) \propto \alpha Z^2 + \beta Z + \gamma$. * For **odd A** , the pairing term is zero. The plot is a single parabola. The minimum represents the single stable isobar. Nuclei higher on the walls undergo β^- or β^+ decay to cascade down to the minimum. * For **even A** , the pairing term splits the plot into two nested parabolas (one for even-even, one for odd-odd). This can lead to multiple stable isobars for a given A (as seen in ^{128}Te and ^{128}Xe), where double- β decay is theoretically possible but single β decay is blocked by energy conservation.

Connections

* **Quantum Mechanics (Uncertainty Principle & Atomic Size):** The radius of an atom is defined by the electron's properties. In the hypothetical case of Muonic Hydrogen (an electron swapped for a muon, $m_\mu \approx 207m_e$), the Bohr radius shrinks by a factor of 207, drastically increasing the overlap of the wavefunction with the nucleus. This leads to the famous "Proton Radius Puzzle," where scattering data and muonic transition data yielded a 7σ discrepancy. * **Fourier Transforms:** The derivation of the form factor $F(\vec{q})$ is a direct application of the Born Approximation from QM scattering theory, analogous to X-ray diffraction in solid-state physics.

Exam / HW Hints

* **Always use Atomic Masses:** When calculating Binding Energy or Q -values, always use m_H (Hydrogen mass) instead of m_p (proton mass) if your table provides atomic masses. This automatically accounts for the Z electron masses to first order. * **SEMF** $Z(Z-1)$ vs Z^2 : Some textbooks approximate the Coulomb term with Z^2 instead of $Z(Z-1)$. $Z(Z-1)$ is strictly correct because a proton does not repel itself, but for large Z , the difference is negligible. Check which convention your professor prefers for numerical grading!

3 Lecture 03 — Nuclear Reactions and Scattering

Overview

This lecture completes the basic catalogue of static nuclear properties. We begin by translating the familiar quantum-mechanical angular momentum algebra into nuclear notation: orbital angular momentum l , intrinsic spin s , single-particle total angular momentum j , and total nuclear spin I . The central physical point is pairing. In even-even nuclei the nucleons pair to $I = 0$, while in odd- A nuclei the spin and parity of the ground state are usually controlled by the unpaired nucleon.

The lecture then introduces parity and isospin as classification quantum numbers. Parity tells us how nuclear wavefunctions and electromagnetic operators behave under $\vec{r} \rightarrow -\vec{r}$, leading directly to selection rules for static multipole moments. Isospin packages the approximate proton-neutron symmetry of the strong interaction into an angular-momentum-like algebra, while emphasizing that isospin lives in an internal space and cannot be added to ordinary spin or orbital angular momentum.

The electromagnetic moments section connects symmetry to measurable nuclear structure. Magnetic dipole moments probe orbital motion, spin, and the composite internal structure of nucleons through their anomalous g -factors. Electric quadrupole moments probe nuclear shape: $Q = 0$ for spherical charge distributions, $Q > 0$ for prolate deformation, and $Q < 0$ for oblate deformation.

The final part of the lecture is problem-oriented. We rederive the form factor of a uniformly charged sphere, then use high-precision molecular mass differences to determine the atomic mass of deuterium and the deuteron binding energy. Independent capture and photodisintegration measurements confirm the same energy scale, preparing the next lecture's model of the deuteron as a bound state in a finite nuclear potential.

Recall from Lecture 02 that electron scattering and form factors give access to the nuclear charge distribution and radius. The empirical fact that nuclear density is approximately constant leads to the radius law $R = r_0 A^{1/3}$ and motivates the liquid-drop picture. We also introduced nuclear masses, binding energies, and the semi-empirical mass formula as a way to organize the dominant contributions to nuclear stability.

In this lecture the focus shifts from bulk nuclear properties to the quantum numbers and moments used to label individual nuclear states:

- angular momentum and nuclear spin,
- parity,
- isospin,
- electromagnetic moments,
- and the deuteron as the first two-nucleon bound state.

Angular Momentum

Classically, angular momentum is the cross product of position and linear momentum:

$$\vec{L} = \vec{r} \times \vec{p}$$

In quantum mechanics the corresponding orbital angular momentum operator is

$$\hat{L} = -i\hbar\vec{r} \times \nabla$$

Its simultaneous eigenstates of L^2 and L_z satisfy

$$L^2|l, m\rangle = \hbar^2 l(l+1)|l, m\rangle$$

$$L_z|l, m\rangle = \hbar m|l, m\rangle$$

where l is the orbital angular momentum quantum number and m is the magnetic quantum number. For a single nucleon, the total single-particle angular momentum is the vector sum of orbital angular momentum and intrinsic spin:

$$\vec{J} = \vec{L} + \vec{S}$$

Spin is an intrinsic property of particles. Protons and neutrons are spin- $\frac{1}{2}$ fermions, so for a nucleon $j = l \pm \frac{1}{2}$.

Spectroscopic Notation

States are labelled by a letter for l followed by a subscript for $j = l \pm \frac{1}{2}$. The letter convention (inherited from atomic spectroscopy) is:

l	Letter	Example states
0	s	$s_{1/2}$
1	p	$p_{1/2}, p_{3/2}$
2	d	$d_{3/2}, d_{5/2}$
3	f	$f_{5/2}, f_{7/2}$
4	g	$g_{7/2}, g_{9/2}$
5	h	$h_{9/2}, h_{11/2}$

In nuclear physics, ground-state configurations rarely require notation beyond h . Pairing tends to cancel the angular momenta of nucleons in time-reversed states, so only an unpaired nucleon usually determines the spin of an odd- A nucleus.

The projection quantum number is defined only after choosing a quantization axis. Experimentally, such an axis is often selected by an external magnetic field, leading to Zeeman splitting of the magnetic substates.

The total angular momentum J for a nucleon is given by the vector sum of the orbital angular momentum L and the spin angular momentum S :

$$\vec{J} = \vec{L} + \vec{S} \quad (15)$$

We define $\vec{I}$ as the total angular momentum of the nucleus, which is the vector sum of the total angular momenta of all the nucleons:

$$\vec{I} = \sum_i \vec{J}_i \quad (16)$$

Thus

$$I^2 = \hbar^2 I(I+1), \quad I_z = \hbar m_I, \quad m_I = -I, -I+1, \dots, I-1, I$$

For even-even nuclei, all protons and neutrons are paired and the ground-state spin is empirically $I = 0$. For odd- A nuclei, the spin is usually set by the unpaired nucleon:

$$\vec{I} = \vec{J}_{\text{unpaired}} \quad (17)$$

where I is the total angular momentum quantum number of the nucleus.

Physical Motivation

The nuclear ground state minimizes the energy. The strong interaction is spin-dependent and favors pairing: two nucleons with equal j but opposite m_j can pair to give $J = 0$. This empirical rule has two immediate consequences:

- **Even–even nuclei** (Z even, N even): all nucleons pair up. *Experimentally, the ground-state spin is always $I = 0$.*
- **Odd-A nuclei**: all nucleons except one pair up. The single unpaired nucleon determines I .

The fact that we never observe ground-state spins of order $A/2$ (which would result if all nucleons added coherently) is direct evidence that pairing is the dominant coupling mode. This is why we do not see very high I values in nuclear tables.

Parity

Parity is the transformation that inverts all spatial coordinates through the origin:

$$\vec{r} \rightarrow -\vec{r}.$$

The parity operator Π acts on a wavefunction $\psi(\vec{r})$:

$$\Pi\psi(\vec{r}) = \psi(-\vec{r}).$$

Applying Π twice returns the original wavefunction, so $\Pi^2 = 1$ and its eigenvalues are $\pi = \pm 1$:

$$\Pi^2\psi(\vec{r}) = \Pi\psi(-\vec{r}) = \psi(\vec{r}). \quad (18)$$

For a single-particle state with orbital angular momentum l , the spherical harmonic $Y_l^m(\theta, \phi)$ acquires a factor $(-1)^l$ under parity, giving:

$$\pi = (-1)^l. \quad (19)$$

States with even l (s, d, g, ...) have *positive* parity; states with odd l (p, f, h, ...) have *negative* parity.

Combined Spin-Parity Notation

Nuclear states are labelled in tables by the combined quantum number I^π , where I is the nuclear spin and $\pi = \pm$ denotes the parity eigenvalue:

$$I^\pi \quad \text{e.g.} \quad 0^+, \quad \frac{3}{2}^-, \quad 2^+, \quad \frac{7}{2}^-.$$

The parity of a nuclear state is determined by the product of the parities of all nucleons. For an even-even nucleus in the $I = 0$ ground state, $\pi = +1$, giving 0^+ .

Isospin

Isospin is an internal quantum number that expresses the approximate symmetry between protons and neutrons under the strong interaction. In this language, the proton and neutron are treated as two charge states of the same particle, the nucleon. We use the convention

$$T_z(p) = +\frac{1}{2}, \quad T_z(n) = -\frac{1}{2}.$$

The total isospin operator of a nucleus is the vector sum of the individual nucleon isospins:

$$\vec{T} = \sum_i \vec{T}_i. \quad (20)$$

The strong interaction is approximately invariant under rotations in this internal isospin space. Therefore nuclei with the same total isospin T , but different T_z , can form isospin multiplets with similar strong-interaction properties. For example, ${}^3\text{He}$ and ${}^3\text{H}$ form an approximate isospin doublet. The symmetry is not exact: electromagnetic interactions and the proton-neutron mass difference split the members of a multiplet.

Mathematically, isospin behaves like angular momentum. The operators T_x , T_y , and T_z obey

$$[T_i, T_j] = i\epsilon_{ijk}T_k \quad (21)$$

where ϵ_{ijk} is the Levi-Civita symbol. The squared total isospin is

$$T^2 = T_x^2 + T_y^2 + T_z^2 \quad (22)$$

and the eigenstates satisfy

$$T^2|T, T_z\rangle = \hbar^2 T(T+1)|T, T_z\rangle \quad (23)$$

and

$$T_z|T, T_z\rangle = \hbar T_z|T, T_z\rangle \quad (24)$$

where $T_z = -T, -T+1, \dots, T$. The key distinction is that isospin is not a spatial angular momentum. It lives in an internal “isospace”, so it cannot be added to ordinary angular momentum or parity.

Electromagnetic moments

Electromagnetic moments provide experimental access to the distribution of charge and current inside the nucleus. The leading non-trivial static moments are the magnetic dipole moment $M1$, which probes angular momentum and current distributions, and the electric quadrupole moment $E2$, which probes departures from spherical symmetry.

Vectors and Pseudovectors

Under the parity transformation $\vec{r} \rightarrow -\vec{r}$, two kinds of vector exist:

- **Polar vector (true vector):** flips sign, e.g. position $\vec{r}$, electric field $\vec{E}$, electric dipole moment $\vec{d}$.

- **Axial vector (pseudovector):** *does not* flip sign, e.g. angular momentum $\vec{L} = \vec{r} \times \vec{p}$ (both $\vec{r}$ and $\vec{p}$ flip, but the cross product brings two sign changes that cancel), magnetic field $\vec{B}$, magnetic moment $\vec{\mu}$.

The key consequence is that the magnetic field $\vec{B}$, being a pseudovector, is *invariant* under parity, while $\vec{E}$ is odd under parity.

Multipole Selection Rules from Parity

The expectation value of any operator $\hat{O}$ in a state of definite parity vanishes unless $\hat{O}$ has *positive* parity:

$$\langle I, \pi | \hat{O} | I, \pi \rangle = 0 \quad \text{if } \hat{O} \text{ has odd parity.} \quad (25)$$

For the multipole expansion, one can show that the parity of the order- L operators is:

$$\pi[E(L)] = (-1)^L, \quad (26)$$

$$\pi[M(L)] = (-1)^{L+1}. \quad (27)$$

Only multipoles with *positive* parity can have non-zero expectation values. This gives the following table of *observable* moments:

Type	Name	Parity	Observable?
$E0$	Electric monopole	+1	Yes (total charge Ze)
$E1$	Electric dipole	−1	No (vanishes by symmetry)
$E2$	Electric quadrupole	+1	Yes
$M0$	Magnetic monopole	−1	No (does not exist in nature)
$M1$	Magnetic dipole	+1	Yes
$M2$	Magnetic quadrupole	−1	No

Thus the leading non-trivial static moments in ordinary nuclei are $M1$ (**magnetic dipole**) and $E2$ (**electric quadrupole**).

Physical significance: The electric dipole vanishes because, for a nucleus, the centre of charge is fixed at the centre of mass (there is no net internal dipole once we work in the centre-of-mass frame).

Magnetic Dipole Moment

Physical Motivation

A classical charged particle orbiting in a circle constitutes a current loop, which generates a magnetic field. Because nucleons carry charge and angular momentum, we expect them to generate magnetic dipole moments. The quantum version of this classical picture gives us the nuclear magnetic moment, which encodes information about both the orbital motion and the intrinsic spin of the nucleons.

Classical derivation. Consider a proton of charge e , mass m_p , moving in a circle of radius r with speed v . The current i and loop area A are:

$$i = \frac{e}{T} = \frac{ev}{2\pi r}, \quad A = \pi r^2.$$

The magnetic dipole moment magnitude is:

$$\mu = iA = \frac{ev}{2\pi r} \cdot \pi r^2 = \frac{evr}{2} = \frac{e}{2m_p} \ell,$$

where $\ell = m_p v r$ is the orbital angular momentum. In vector form:

$$\vec{\mu}_\ell = \frac{e}{2m_p} \vec{L}. \quad (28)$$

Quantum mechanical version. Promoting L to an operator and introducing a dimensionless g -factor to account for internal structure gives

$$\vec{\mu} = g_l \frac{e}{2m_p} \vec{L} = g_l \mu_N \vec{L}/\hbar, \quad (29)$$

where $\mu_N \equiv e\hbar/(2m_p)$ is the *nuclear magneton*. For a complete nucleus, the measured magnetic moment is conventionally written in terms of the nuclear spin I and a nuclear g -factor,

$$\mu = g \mu_N I, \quad (30)$$

where g depends on the proton and neutron configuration. This moment receives contributions from both orbital motion and intrinsic spin.

The **Bohr magneton** (unit for atomic/electron moments):

$$\mu_B = \frac{e\hbar}{2m_e} \approx 5.788 \times 10^{-5} \text{ eV/T}. \quad (31)$$

The **nuclear magneton** (unit for nuclear moments; smaller by $m_e/m_p \approx 1/1836$):

$$\mu_N = \frac{e\hbar}{2m_p} \approx 3.152 \times 10^{-8} \text{ eV/T}. \quad (32)$$

Physical significance: Because $\mu_N \ll \mu_B$, nuclear magnetic moments are about 1836 times weaker than electron magnetic moments. This is why spin-resonance experiments on nuclei require much smaller energy splittings than corresponding electron-spin experiments.

Orbital g -factors

The orbital magnetic moment of a nucleon is:

$$\vec{\mu}_l = g_l \mu_N \frac{\vec{L}}{\hbar}, \quad (33)$$

with:

$$g_l(\text{proton}) = 1, \quad g_l(\text{neutron}) = 0.$$

The proton's $g_l = 1$ follows from its charge $+e$; the neutron has $g_l = 0$ because its total charge is zero.

Spin g -factors

The spin magnetic moment is:

$$\vec{\mu}_s = g_s \mu_N \frac{\vec{S}}{\hbar}. \quad (34)$$

From the Dirac equation, a *point-like* spin- $\frac{1}{2}$ particle satisfies $g_s = 2$. The *measured* values for nucleons, however, deviate strongly:

$$\boxed{g_s(p) = 5.586, \quad g_s(n) = -3.826.} \quad (35)$$

Physical significance: These values differ dramatically from $g_s = 2$, proving that protons and neutrons are *not* point-like particles. They are composite objects built from quarks and gluons. In fact, $\sim 99\%$ of the proton mass comes from the gluon field binding energy (QCD confinement), not from the quark rest masses. The anomalous g -factors are a direct measurement of this internal structure.

The measured magnetic moments of the proton and neutron are:

$$\mu_p \approx +2.793 \mu_N, \quad \mu_n \approx -1.913 \mu_N.$$

The Muon $g - 2$ Problem

The Dirac prediction $g_s \approx 2$ is corrected by quantum field theory (QED) to:

$$g = 2 \left(1 + \frac{\alpha}{2\pi} + \dots \right).$$

For the *electron*, this structure is confirmed to extraordinary precision. For the *muon*, dedicated $g-2$ experiments measure the precession of muons in a storage ring and compare the anomalous magnetic moment $a_\mu = (g - 2)/2$ with Standard Model calculations. The muon case is an important example of how high-precision moment measurements can test physics beyond the simplest Dirac picture.

Electric Quadrupole Moment

Physical Motivation

The monopole moment tells us the total charge, and the dipole moment vanishes for a nucleus in its centre-of-mass frame. The next electric moment that can reveal nuclear shape is therefore the quadrupole moment. It answers a simple geometric question: is the charge distribution spherical, stretched along one axis, or flattened perpendicular to that axis?

The electric quadrupole moment is

$$\boxed{Q = \int \rho(\vec{r}) (3z^2 - r^2) d^3r} \quad (36)$$

where $\rho(\vec{r})$ is the charge density, z is measured along the chosen symmetry axis, and $r^2 = x^2 + y^2 + z^2$.

Physical significance: A nonzero Q is direct evidence that the nuclear charge distribution is not spherical.

For a spherically symmetric distribution,

$$\langle x^2 \rangle = \langle y^2 \rangle = \langle z^2 \rangle, \quad \langle r^2 \rangle = 3\langle z^2 \rangle,$$

so the integrand averages to zero and

$$Q_{\text{spherical}} = 0.$$

For a prolate nucleus elongated along the z axis, z^2 is large compared with $x^2 + y^2$, so $Q > 0$. For an oblate nucleus flattened along z , z^2 is comparatively small and $Q < 0$.

For a typical single-particle deformation one expects the scale

$$\boxed{|Q| \sim eR^2 \sim er_0^2 A^{2/3}.} \quad (37)$$

Physical significance: Quadrupole moments of this order indicate modest surface deformation, while values much larger than this signal collective deformation of the entire nucleus.

This closes the survey of basic nuclear properties: mass and charge distributions, radius, binding energy, the semi-empirical mass formula, angular momentum, parity, isospin, magnetic dipole moments, and electric quadrupole moments.

Worked Problem: Uniform-Sphere Form Factor

Physical Motivation

In electron scattering, the form factor is the Fourier transform of the charge distribution. It tells us which momentum transfers see the nucleus as point-like and which begin to resolve its finite size. The simplest useful model is a sphere of uniform charge density.

Let

$$\rho(r) = \begin{cases} \rho_0, & r \leq R, \\ 0, & r > R. \end{cases}$$

The unnormalized Fourier transform is

$$\tilde{F}(q) = \int \rho(\vec{r}) e^{i\vec{q} \cdot \vec{r}} d^3r.$$

Choose the z axis along $\vec{q}$, so $\vec{q} \cdot \vec{r} = qr \cos \theta$. Then

$$\begin{aligned} \tilde{F}(q) &= \int_0^R \int_0^\pi \int_0^{2\pi} \rho_0 e^{iqr \cos \theta} r^2 \sin \theta d\phi d\theta dr \\ &= 2\pi \rho_0 \int_0^R r^2 \left(\int_0^\pi e^{iqr \cos \theta} \sin \theta d\theta \right) dr. \end{aligned} \quad (38)$$

With $u = \cos \theta$,

$$\begin{aligned} \int_0^\pi e^{iqr \cos \theta} \sin \theta d\theta &= \int_{-1}^1 e^{iqr u} du \\ &= \frac{e^{iqr} - e^{-iqr}}{iqr} = \frac{2 \sin(qr)}{qr}. \end{aligned}$$

Therefore

$$\begin{aligned}\tilde{F}(q) &= \frac{4\pi\rho_0}{q} \int_0^R r \sin(qr) dr \\ &= \frac{4\pi\rho_0}{q} \left[-\frac{r \cos(qr)}{q} + \frac{\sin(qr)}{q^2} \right]_0^R \\ &= \frac{4\pi\rho_0}{q^3} [\sin(qR) - qR \cos(qR)].\end{aligned}$$

The normalized form factor divides by the total charge

$$\tilde{F}(0) = \int \rho(\vec{r}) d^3r = \frac{4\pi}{3} \rho_0 R^3.$$

Thus

$$F(q) = 3 \frac{\sin(qR) - qR \cos(qR)}{(qR)^3}. \quad (39)$$

Physical significance: The limit $F(0) = 1$ is the normalization condition; the first zero of $F(q)$ gives an experimental handle on the nuclear radius R .

To check the $q \rightarrow 0$ limit explicitly, write $x = qR$ and expand:

$$\sin x - x \cos x = \left(x - \frac{x^3}{6} + \cdots \right) - x \left(1 - \frac{x^2}{2} + \cdots \right) = \frac{x^3}{3} + \mathcal{O}(x^5).$$

Hence

$$F(q) \xrightarrow{q \rightarrow 0} 3 \frac{x^3/3}{x^3} = 1,$$

as required.

Worked Problem: Measuring the Deuterium Atomic Mass

Physical Motivation

Absolute mass measurements are hard because the masses are large on the nuclear-energy scale. Nuclear binding energies are only MeV, while atomic masses are of order GeV. The practical strategy is therefore to measure small mass differences between very similar molecular ions; many systematic errors then cancel.

For a simple mass spectrometer with crossed electric and magnetic fields, the measured quantity is essentially

$$\frac{m}{q} = \frac{BB'R}{\mathcal{E}}, \quad (40)$$

where $\mathcal{E}$ is the electric field in the velocity selector, B' is the magnetic field in the analyzer, and R is the orbit radius.

Physical significance: Measuring nearby mass peaks on the same instrument makes the difference much more precise than either absolute mass alone.

The lecture used molecular mass differences to infer the atomic mass of deuterium, m_D . In the first measurement,

$$M(\text{C}_6\text{H}_{12}) - M(\text{C}_6\text{D}_6) = 9.289710(24) \times 10^{-3} \text{ u}.$$

Neglecting molecular electronic binding energies, which are eV-scale rather than MeV-scale,

$$\begin{aligned} M(\text{C}_6\text{H}_{12}) - M(\text{C}_6\text{D}_6) &= (6m_C + 12m_H) - (6m_C + 6m_D) \\ &= 12m_H - 6m_D. \end{aligned}$$

Solving for m_D gives

$$\begin{aligned} m_D &= 2m_H - \frac{1}{6}[M(\text{C}_6\text{H}_{12}) - M(\text{C}_6\text{D}_6)] \\ &= 2(1.00782503237 \text{ u}) - \frac{9.289710 \times 10^{-3} \text{ u}}{6} \\ &= 2.01410178 \text{ u}. \end{aligned}$$

The uncertainty is divided by the exact factor 6:

$$\sigma_1 = \frac{2.4 \times 10^{-8} \text{ u}}{6} = 4.0 \times 10^{-9} \text{ u}.$$

Thus

$$\boxed{m_D^{(1)} = 2.014101780(4) \text{ u}.} \quad (41)$$

Physical significance: The relevant uncertainty is in the ninth decimal place of an atomic mass unit, which is why the binding energy can be extracted on the keV scale.

A second molecular comparison in the lecture gave

$$\boxed{m_D^{(2)} = 2.014101771(15) \text{ u}.} \quad (42)$$

Physical significance: The two values differ by about 10^{-8} u, comparable to their combined uncertainty, so the measurements are statistically consistent.

Combining the two values by inverse-variance weighting gives approximately

$$\boxed{m_D = 2.014101779(4) \text{ u}.} \quad (43)$$

Physical significance: Independent measurements agreeing within uncertainty are what make the subsequent binding-energy extraction credible.

Binding Energy of the Deuteron

Physical Motivation

The deuteron is the lightest bound nucleus: one proton plus one neutron. Its binding energy is therefore the cleanest first test of whether the nuclear force can actually bind nucleons. If the deuteron were not bound, ordinary nuclear matter would be radically different.

Using atomic masses is convenient because the electron in atomic hydrogen cancels the electron in atomic deuterium. Therefore

$$\boxed{B_d = (m_H + m_n - m_D)c^2.} \quad (44)$$

Physical significance: A positive B_d means that the deuteron lies below the free proton-neutron threshold and is a stable bound state.

Substitute

$$m_H = 1.00782503237 \text{ u}, \quad m_n = 1.00866491588 \text{ u}, \quad m_D = 2.014101779 \text{ u},$$

and $1 \text{ u} = 931.494 \text{ MeV}/c^2$:

$$\begin{aligned} B_d &= (1.00782503237 + 1.00866491588 - 2.014101779)(931.494 \text{ MeV}) \\ &= 0.00238816925 \times 931.494 \text{ MeV} \\ &= 2.22456 \text{ MeV}. \end{aligned}$$

Thus

$$\boxed{B_d \simeq 2.2246 \text{ MeV}.} \quad (45)$$

Physical significance: The deuteron is bound, but weakly compared with typical nuclear binding energies of about 8 MeV per nucleon.

Independent Experimental Checks

The same binding energy can be checked without relying only on mass differences.

In radiative capture,

$$p + n \rightarrow d + \gamma,$$

the photon carries away almost all the released binding energy, with a small correction from recoil:

$$\boxed{E_\gamma \approx B_d.} \quad (46)$$

Physical significance: The emitted photon is a direct calorimeter for the energy gained when the proton and neutron fall into the bound deuteron state.

The lecture quoted

$$B_d = 2.224589(2) \text{ MeV}$$

from such a capture measurement.

Conversely, in photodisintegration,

$$\gamma + d \rightarrow p + n,$$

the threshold photon energy is approximately the binding energy:

$$\boxed{E_{\gamma,\text{thr}} \approx B_d.} \quad (47)$$

Physical significance: The threshold is the minimum energy needed to lift the deuteron out of the bound-state potential well.

The lecture quoted the less precise but consistent value

$$B_d \approx 2.224 \pm 0.020 \text{ MeV}.$$

The agreement among mass spectroscopy, radiative capture, and photodisintegration confirms that the deuteron binding energy is a real physical scale, not an artifact of one measurement method.

Looking Ahead: Modeling the Deuteron

Physical Motivation

Once the binding energy is known, the next question is whether a simple potential model can reproduce it. The deuteron is the simplest arena for testing a quantum-mechanical model of the nuclear force.

The next step is to approximate the proton-neutron interaction by a finite, central potential $V(r)$ and solve the three-dimensional Schrödinger equation. Separating variables into radial and angular parts produces a radial equation with a centrifugal term:

$$\boxed{-\frac{\hbar^2}{2\mu} \frac{d^2 u_l}{dr^2} + \left[V(r) + \frac{\hbar^2 l(l+1)}{2\mu r^2} \right] u_l(r) = E u_l(r),} \quad (48)$$

where $\mu = m_p m_n / (m_p + m_n)$ is the reduced mass, l is the orbital angular momentum, and $u_l(r)$ is the reduced radial wavefunction.

Physical significance: Bound-state solutions of this equation are the mathematical statement that the nuclear force can hold a proton and neutron together; Lecture 04 will use this structure to constrain the depth and range of the nuclear potential.

Further Reading

Recommended Textbooks

- Kenneth S. Krane, *Introductory Nuclear Physics*. Most relevant sections: Chapter 2.5–2.6 for angular momentum and parity; Chapter 3.3–3.5 for binding energy, nuclear spin/parity, and electromagnetic moments; Chapter 4.1 for the deuteron; Chapter 12.3 for isospin in reactions.
- Carlos A. Bertulani, *Nuclear Physics in a Nutshell*. Most relevant sections: Chapter 1.5–1.8 for parity, isospin, and nucleon magnetic moments; Chapter 2.2–2.6 for multipoles and the deuteron; Chapter 4.4–4.7 for nuclear angular momentum, multipole moments, magnetic dipole moments, and electric quadrupole moments; Appendices A–C for angular momentum and symmetries.
- Samuel S. M. Wong, *Introductory Nuclear Physics*. Read the chapters on nucleon structure, nuclear force and two-nucleon systems, and the nuclear shell model for a more formal treatment of isospin, the deuteron, and spin-parity assignments.
- K. Heyde, *Basic Ideas and Concepts in Nuclear Physics*. Useful as a conceptual companion for nuclear properties, electromagnetic moments, and the structure of light nuclei.

Web Resources

- [IAEA LiveChart of Nuclides](#): interactive evaluated nuclear data, including spins, parities, decay data, radii, and nuclear moments.
- [NNDC NuDat 3](#): searchable nuclear levels, half-lives, spin-parity assignments, gamma transitions, and reaction data.

- [NIST Atomic Weights and Isotopic Compositions](#): reference values for isotope masses used in binding-energy and Q -value calculations.
- [NIST Atomic Data for Hydrogen](#): quick reference for hydrogen and deuterium isotope masses, spin, and magnetic moment.
- [NNDC table of nuclear magnetic dipole and electric quadrupole moments](#): useful for comparing measured moments with the qualitative shape arguments in this lecture.

4 Lecture 04 — Deuteron Ground State and the Nuclear Shell Model

Overview

This lecture has two parts. The first solves the quantum mechanics of the deuteron: we determine its ground-state angular momentum by combining parity and magnetic-moment arguments, discover that the ground state is a superposition of $l = 0$ and $l = 2$ components (proving the nuclear force has a tensor part), and solve the Schrödinger equation for a square-well potential to extract the potential depth $V_0 \approx 35 \text{ MeV}$. The second part introduces the nuclear shell model: we motivate the existence of nuclear magic numbers, justify why independent-particle motion is not unreasonable, try several model potentials and fail, and finally show that a spin-orbit coupling term $V_{\text{so}} \propto \vec{l} \cdot \vec{s}$ reorganizes the levels so as to reproduce all observed magic numbers.

5 Part I — Solving the Deuteron

5.1 Determining the Orbital Angular Momentum

Physical Motivation

We know from Lecture 03 that the deuteron binding energy is $B_d \approx 2.22 \text{ MeV}$ and that its spin-parity is $I^\pi = 1^+$. We now ask: what orbital angular momentum l does the proton–neutron pair carry? The answer is not trivial, and it turns out to be more interesting than a simple integer.

The measured spin-parity $I^\pi = 1^+$ constrains the possible l values. Parity of the spatial part of the wavefunction is $(-1)^l$. For the nuclear state to have $\pi = +1$ we need $(-1)^l = +1$, so l must be *even*: $l = 0, 2, 4, \dots$. Including the spin of the nucleon pair $S = 0$ (singlet) or $S = 1$ (triplet) and requiring the total $J = 1$ reduces the candidates further:

- $l = 0, S = 1: J = 1, \pi = +1 \quad \checkmark$
- $l = 2, S = 1: J = 1, 2, 3$ (all have $\pi = +1$), $J = 1$ is possible $\checkmark$
- $l = 1, S = 0: J = 1, \pi = -1 \quad \times$ (wrong parity)
- $l = 1, S = 1: J = 0, 1, 2, \pi = -1 \quad \times$ (wrong parity)

So **two choices survive**: $l = 0$ or $l = 2$ (or a mixture).

Using the Magnetic Moment to Distinguish

Prediction for $l = 0$ (pure S-wave). If $l = 0$ there is no orbital contribution and the magnetic moment comes entirely from the spins. With both nucleons in the triplet $S = 1$ state:

$$\mu_{l=0} = \frac{1}{2} [g_s(p) + g_s(n)] \mu_N = \frac{1}{2} (5.586 - 3.826) \mu_N \approx 0.879 \mu_N. \quad (49)$$

The measured value is

$$\mu_d = 0.8574 \mu_N. \quad (50)$$

These are close but do *not* agree — the discrepancy is well outside experimental uncertainty.

Prediction for $l = 2$ (pure D-wave). A more involved calculation gives $\mu_{l=2} \approx 0.31 \mu_N$, which is even further from the measured value.

Conclusion: a mixed S–D ground state. Neither pure $l = 0$ nor pure $l = 2$ fits. The resolution is that the deuteron ground state is a *superposition*:

$$\psi_d = A \psi_{S\text{-wave}} + B \psi_{D\text{-wave}}, \quad (51)$$

with normalization $|A|^2 + |B|^2 = 1$. Matching the measured μ_d fixes the D-wave probability:

$$\boxed{|B|^2 \equiv P_D \approx 0.04.} \quad (52)$$

Physical significance: About 4% of the time the proton and neutron are in an $l = 2$ state. Because l is not fixed, l is *not a good quantum number* for the deuteron. A central potential cannot mix $l = 0$ and $l = 2$; therefore the nuclear force contains a **non-central (tensor) component**. The tensor force is the dominant deviation from a simple central interaction.

5.2 Square-Well Model for the Deuteron

Physical Motivation

With $P_D \approx 4\%$ being small, we can work to good approximation in the $l = 0$ sector alone. The goal is to relate the measured binding energy $B_d \approx 2.22$ MeV and the known nuclear radius $R \approx 2$ fm to the depth V_0 of the nuclear potential. The simplest possible potential that is short-ranged and attractive is the square well.

Reduced-mass setup. Working in the centre-of-mass frame, the two-body problem reduces to a one-body problem with the reduced mass

$$\mu = \frac{m_p m_n}{m_p + m_n} \approx \frac{m_N}{2}, \quad (53)$$

and the relative coordinate $\vec{r}$. The wavefunction factorises as $\psi(\vec{r}) = R_{nl}(r)Y_l^m(\theta, \phi)$. Setting $u(r) = rR(r)$ and $l = 0$, the radial Schrödinger equation becomes

$$-\frac{\hbar^2}{2\mu} \frac{d^2 u}{dr^2} + V(r)u(r) = Eu(r). \quad (54)$$

For a bound state $E = -B < 0$.

Square-well potential.

$$V(r) = \begin{cases} -V_0 & r < R, \\ 0 & r \geq R. \end{cases} \quad (55)$$

Solutions in each region. Define

$$k_1 = \sqrt{\frac{2\mu(V_0 - B)}{\hbar^2}}, \quad k_2 = \sqrt{\frac{2\mu B}{\hbar^2}}. \quad (56)$$

- **Inside** ($r < R$): equation $u'' + k_1^2 u = 0$; general solution $A \sin k_1 r + C \cos k_1 r$. The boundary condition $u(0) = 0$ (so that $\psi(0) = u(0)/r$ stays finite) forces $C = 0$:

$$u_{\text{in}}(r) = A \sin k_1 r. \quad (57)$$

- **Outside** ($r > R$): equation $u'' - k_2^2 u = 0$; general solution $De^{-k_2 r} + Fe^{+k_2 r}$. Normalizability forces $F = 0$:

$$u_{\text{out}}(r) = De^{-k_2 r}. \quad (58)$$

Matching boundary conditions at $r = R$. Continuity of u and u' :

$$A \sin k_1 R = De^{-k_2 R}, \quad (59)$$

$$Ak_1 \cos k_1 R = -k_2 De^{-k_2 R}. \quad (60)$$

Dividing eliminates A and D :

$$\boxed{k_1 \cot(k_1 R) = -k_2.} \quad (61)$$

Physical significance: This transcendental equation, together with the definitions of k_1 and k_2 (which depend on V_0 , B , μ), uniquely determines V_0 once B_d and R are known from experiment.

Numerical result. Inserting $B_d = 2.22 \text{ MeV}$, $R \approx 2 \text{ fm}$, $\mu \approx m_N/2$:

$$\boxed{V_0 \approx 35 \text{ MeV}.} \quad (62)$$

The deuteron sits near the very edge of the well: $k_1 R \approx \pi/2$, meaning the wavefunction inside is just reaching its first quarter-oscillation when it meets the wall.

Minimum Potential Depth for a Bound State

A bound state requires $k_1 R \geq \pi/2$, i.e.

$$V_{0,\text{min}} = \frac{\pi^2 \hbar^2}{8\mu R^2} \approx 25 \text{ MeV}. \quad (63)$$

Physical significance: The actual depth $V_0 \approx 35 \text{ MeV}$ is only 40% above the minimum needed. The deuteron is barely bound. Had the nuclear force been slightly weaker, no bound state of the two nucleons would exist, and nuclear chemistry as we know it would be impossible.

6 Part II — The Nuclear Shell Model

6.1 Magic Numbers and Empirical Evidence

Physical Motivation

Atoms are stable when electron shells are complete (noble gases). By analogy, nuclei might be especially stable when their proton or neutron numbers fill particular “shells.” The numbers at which this happens are called **magic numbers**:

$$\boxed{2, \quad 8, \quad 20, \quad 28, \quad 50, \quad 82, \quad 126.} \quad (64)$$

Several independent experimental observations converge on the same numbers:

1. **Number of stable isotopes/isotones** — magic- N or magic- Z nuclides have unusually many stable species. For example, $N = 28$ has 5–6 stable isotones; neighbouring N values have only 1–2.
2. **Alpha-decay energies** — the kinetic energy released in α decay of nuclei with $N = 126$ (a filled neutron shell) drops sharply as N crosses 126, then rises again.
3. **Neutron-capture cross sections** — nuclei at magic N have anomalously *small* cross sections: the full shell resists adding one more neutron.
4. **Nuclear radii** — the empirical $R = r_0 A^{1/3}$ trend shows kinks at magic numbers.

The pattern unmistakably echoes atomic shell closure.

6.2 Why Independent-Particle Motion is Plausible

Physical Motivation

At first sight, treating nucleons as free particles in a mean potential seems absurd: the nucleus is a dense system where every nucleon interacts strongly with its neighbours. Yet the Pauli exclusion principle saves the picture.

In the ground state, all orbitals up to the Fermi energy are occupied. For two nucleons to scatter, one must transfer enough energy to the other to kick it into an unoccupied orbital above the Fermi level. But *all nearby states are already full*. The energy required is $\sim E_F \approx 33$ MeV, far larger than the thermal energy scale. As a result, nucleons effectively *pass through each other without scattering*, and each nucleon moves in a slowly varying mean-field potential $V(r)$ created by all the others. This is the **independent-particle approximation**.

6.3 Mean-Field Potentials and their Failures

Infinite square well. Solving $-\frac{\hbar^2}{2m}\nabla^2\psi = E\psi$ inside a sphere of radius R gives energy levels labelled by (n_r, l) . Filling them with $2(2l+1)$ nucleons each (factor 2 for spin) produces cumulative occupations that hit shell gaps at

$$2, 8, 20, 40, 58, 92, \dots$$

The first three (2, 8, 20) agree with experiment; beyond 20 the model fails.

Harmonic oscillator. Energy levels $E_N = \hbar\omega(N+3/2)$ with degeneracy $(N+1)(N+2)/2$ per oscillator quantum $N = 2n_r + l$. Shell closures occur at

$$2, 8, 20, 40, 70, \dots$$

Again, only the first three magic numbers are reproduced.

Woods–Saxon potential. A more realistic rounded well,

$$V(r) = \frac{-V_0}{1 + \exp\left(\frac{r-R}{a}\right)}, \quad (65)$$

where $R \approx r_0 A^{1/3}$ is the nuclear radius and $a \approx 0.5$ fm is the skin thickness, gives a softer level sequence. It agrees better than the square well with observed radii and separation energies, but still fails to reproduce magic numbers beyond 20.

Physical significance: The parameter a in the Woods–Saxon potential is the same skin thickness that appears in the charge-density profiles measured by electron scattering in Lecture 02, confirming self-consistency.

6.4 Spin-Orbit Coupling Solves the Problem

Physical Motivation

In atomic physics, the electron’s spin couples to the magnetic field generated by its orbital motion, splitting each l -level into $j = l + \frac{1}{2}$ and $j = l - \frac{1}{2}$ sub-levels. Maria Goeppert Mayer and J. Hans D. Jensen showed (1949, Nobel Prize 1963) that adding a spin-orbit term to the nuclear mean field,

$$V_{\text{so}}(r) = -V_{\text{ls}} f(r) \vec{l} \cdot \vec{s}, \quad (66)$$

with $V_{\text{ls}} > 0$ and $f(r) \propto -dV_{\text{WS}}/dr$, generates the correct magic numbers. The sign is important: in *nuclei* the $j = l + \frac{1}{2}$ state is pushed *down* (more bound), opposite to the atomic case.

Evaluating $\vec{l} \cdot \vec{s}$. Using $\vec{j} = \vec{l} + \vec{s}$ and squaring:

$$\vec{l} \cdot \vec{s} = \frac{1}{2}(\vec{j}^2 - \vec{l}^2 - \vec{s}^2). \quad (67)$$

In a state $|j, l, s\rangle$ with $s = \frac{1}{2}$:

$$\langle \vec{l} \cdot \vec{s} \rangle = \begin{cases} +\frac{\hbar^2 l}{2} & j = l + \frac{1}{2}, \\ -\frac{\hbar^2 (l+1)}{2} & j = l - \frac{1}{2}. \end{cases} \quad (68)$$

The energy splitting between the two sub-levels is

$$\Delta E = V_{\text{ls}} f_0 \frac{\hbar^2}{2} (2l+1), \quad (69)$$

which *grows with l* . For large l (e.g. $l = 4$, the g -shell), the splitting is large enough to push a level from one major oscillator shell into the one below it, rearranging the cumulative level occupations and producing the correct gaps at 28, 50, 82, and 126.

Physical significance: Before Mayer and Jensen, the failure at 28, 50, 82, 126 was an embarrassment. Adding a single physically motivated term — the spin-orbit interaction — solved the entire problem at once.

6.5 The Extreme Independent-Particle Model (EIPM)

Physical Motivation

Once the shell structure is established, predicting the ground-state spin and parity of a nucleus becomes a counting exercise. For simplicity we use the **Extreme Independent-Particle Model**: even- A nuclei have $I = 0$ (all pairs cancel), and for odd- A nuclei *only the last unpaired nucleon determines I^π* .

Rule. Fill the proton and neutron shells separately up to the A -th nucleon. The last unpaired nucleon occupies a state with orbital quantum number l and total angular momentum j . Then:

$$\boxed{I^\pi = j^\pi, \quad \pi = (-1)^l.} \quad (70)$$

Example: ^{17}O ($Z = 8$, $N = 9$). The eight protons fill $1s_{1/2}$, $1p_{3/2}$, $1p_{1/2}$ completely; since $Z = 8$ is magic, protons contribute $I = 0$. The nine neutrons fill the same sequence plus one extra neutron that must go into the $1d_{5/2}$ level:

Level	Capacity	Cumulative N
$1s_{1/2}$	2	2
$1p_{3/2}$	4	6
$1p_{1/2}$	2	8
$1d_{5/2}$	6	14

The 9th neutron sits in $1d_{5/2}$, so $j = \frac{5}{2}$, $l = 2$, $\pi = +1$:

$$I^\pi(^{17}\text{O}) = \frac{5}{2}^+. \quad \checkmark \quad (71)$$

(The experimental value is indeed $\frac{5}{2}^+$.)

Example: ^{15}O ($Z = 8$, $N = 7$). Seven neutrons fill $1s_{1/2}$ (2), $1p_{3/2}$ (4), and one of the $1p_{1/2}$ states. The last neutron is in $1p_{1/2}$: $j = \frac{1}{2}$, $l = 1$, $\pi = -1$:

$$I^\pi(^{15}\text{O}) = \frac{1}{2}^-. \quad \checkmark \quad (72)$$

Successes and Limitations

The EIPM reproduces the *spin-parity* of most odd- A ground states correctly. Magnetic moments are only reproduced qualitatively (the predicted values lie between the Schmidt lines; real nuclei cluster near the lines but are not on them). Electric quadrupole moments are sometimes off by large factors, signalling that collective deformation beyond the single-particle picture is present.

Summary of Key Results

1. Deuteron ground state: $\psi_d = A\psi_S + B\psi_D$, $P_D = |B|^2 \approx 4\%$. The $l = 0$ and $l = 2$ admixture proves the nuclear force has a tensor component.
2. Square-well model: boundary condition $k_1 \cot(k_1 R) = -k_2$ gives $V_0 \approx 35 \text{ MeV}$; minimum for a bound state is $V_{0,\min} \approx 25 \text{ MeV}$.
3. Magic numbers 2, 8, 20 follow from both the infinite-square-well and the harmonic-oscillator potential. Reproducing 28, 50, 82, 126 requires spin-orbit coupling $V_{\text{so}} \propto -\vec{l} \cdot \vec{s}$.
4. EIPM prediction: for odd- A , $I^\pi = j^\pi$ of the last unpaired nucleon, with $\pi = (-1)^l$.

7 Lecture 05 — Radioactive Decay, Activity, and Radiation Dose

Overview

This lecture opens the second part of the course. The first part treated the nucleus as a bound quantum system: nuclear sizes, binding energies, scattering, the deuteron, and shell structure. We now turn to the processes by which an unstable nucleus changes in time. The central idea is simple but powerful: radioactive decay is a *statistical quantum process* in which a nucleus loses energy by moving to a more favourable nuclear state, often by becoming a different nucleus.

The lecture has four main goals. First, we classify the most important decay modes: γ decay, internal conversion, α decay, fission, β decay, electron capture, and delayed particle emission. Second, we quantify radioactivity using the decay constant, activity, half-life, and mean lifetime. Third, we study decay branches and decay chains, where a radioactive parent produces radioactive daughters. Finally, we introduce the language used in radiation safety: exposure, absorbed dose, equivalent dose, and the practical principles ALARA and LNT.

7.1 What Counts as Radioactive Decay?

Physical Motivation

A nucleus can be energetically unstable even when it is a perfectly well-defined bound quantum state. It may sit in a local minimum: classically it looks stable, but quantum mechanically there can be a small probability per unit time to escape to a lower-energy configuration. This is why a nucleus can have a lifetime ranging from less than a nanosecond to longer than the age of the Universe.

Physical Concept 7.1: Radioactive decay

Radioactive decay is a process in which a nucleus in its ground state or in an excited state releases energy and moves to a lower-energy nuclear configuration. The final configuration may be the same nucleus in a lower internal state, or a different nucleus.

The energy scale is set by nuclear binding energies. In most ordinary nuclear decays the released energy is of order

$$Q \sim 10^2 \text{ keV} - 10 \text{ MeV}. \quad (73)$$

Physical significance: This is much larger than atomic transition energies, but much smaller than the rest energy of a nucleus. Fission is the main exception: a single fission event releases roughly 200 MeV.

Because decay is probabilistic, we should not ask, “When will this particular nucleus decay?” as if the answer were deterministic. The meaningful question is: *what is the probability per unit time that it decays?* That question leads directly to the decay constant λ in Section 7.4.

7.2 Decay Modes

Physical Motivation

The nucleus must get rid of energy while preserving the conservation laws: energy, momentum, angular momentum, electric charge, baryon number, and lepton number where relevant. Different decay modes are different ways of satisfying those constraints. Some modes keep the identity of the nucleus fixed; others change Z , A , or both.

Gamma decay. In γ decay an excited nucleus emits a high-energy photon and remains the same nuclide:



The photon carries almost all the transition energy, but not exactly all of it: the daughter nucleus must recoil so that momentum is conserved.

Physical significance: A laboratory “gamma source” is usually not a source whose nuclei sit for macroscopic times in excited states. Excited nuclear states typically live for very short times, often nanoseconds or less. The gamma photon is usually produced after some earlier decay, such as β decay, leaves the daughter nucleus excited.

Internal conversion. An excited nucleus can also give its energy directly to an atomic electron, which is then ejected:



The ion charge reminds us that an atomic electron has been removed. In practice one often focuses on the nuclear transition and treats the atomic rearrangement separately.

Physical significance: Internal conversion competes with gamma emission because both start from the same excited nuclear state. Which channel dominates depends on the transition energy, multipolarity, and the electron wavefunction near the nucleus.

Alpha decay. In α decay a heavy nucleus emits a helium-4 nucleus:



The mass number decreases by 4 and the proton number decreases by 2. Although the α particle is strongly bound, it must typically tunnel through the Coulomb barrier to escape the parent nucleus.

Physical significance: Alpha decay is a clean example of a nucleus being energetically allowed to move to a lower state, while still being classically trapped for a long time. The lifetime is extremely sensitive to the barrier width and to the decay energy.

Fission. In fission a very heavy nucleus splits into two medium-mass fragments, usually with extra neutrons and gamma rays:



Here $\bar{\nu}$ is the average number of emitted neutrons. The notation is schematic: real fission has many possible fragment pairs and neutron multiplicities.

Physical significance: Fission releases an unusually large energy, about 200 MeV per event, because the fragments lie closer to the maximum of binding energy per nucleon than the original very heavy nucleus.

Beta-minus decay. In β^- decay, a neutron inside the nucleus turns into a proton:

$$\boxed{{}_Z^AX \rightarrow {}_{Z+1}^AY + e^- + \bar{\nu}_e.} \quad (78)$$

The emitted electron preserves electric charge, and the electron antineutrino preserves lepton number. At the nucleon level,

$$n \rightarrow p + e^- + \bar{\nu}_e. \quad (79)$$

Physical significance: Because the final state contains three bodies (the daughter nucleus, electron, and antineutrino), the electron energy is not a single sharp value. It forms a continuous spectrum from near zero up to the endpoint energy.

Beta-plus decay. In β^+ decay, a proton inside the nucleus turns into a neutron:

$$\boxed{{}_Z^AX \rightarrow {}_{Z-1}^AY + e^+ + \nu_e.} \quad (80)$$

The positron will usually annihilate with an electron, producing two photons:

$$e^+ + e^- \rightarrow \gamma + \gamma, \quad E_\gamma = 511 \text{ keV}. \quad (81)$$

The two photons are emitted nearly back-to-back in the centre-of-mass frame.

Physical significance: This back-to-back photon signature is the basis of positron emission tomography (PET): a positron-emitting isotope is placed in the body, and detectors reconstruct where the annihilations occurred.

Electron capture. Electron capture competes with β^+ decay. Instead of emitting a positron, the nucleus captures one of its atomic electrons:

$$\boxed{{}_Z^AX + e^- \rightarrow {}_{Z-1}^AY + \nu_e.} \quad (82)$$

At the nucleon level, a proton and electron combine into a neutron and a neutrino.

Physical significance: Electron capture and β^+ decay both reduce Z by one while keeping A fixed. They are therefore competing channels for proton-rich nuclei.

Delayed proton and neutron emission. Some nuclei are produced in excited states above the particle-emission threshold. They can then emit a proton or neutron:

$${}_Z^AX^* \rightarrow {}_{Z-1}^{A-1}Y + p, \quad (83)$$

$${}_Z^AX^* \rightarrow {}_Z^{A-1}Y + n. \quad (84)$$

These decays are less common in naturally occurring radioactivity, but they are important near the proton and neutron drip lines and in accelerator-produced nuclei.

Physical significance: Particle emission reminds us that “decay mode” is not only about the parent nucleus. It also depends on which daughter states are energetically accessible.

7.3 Branches, Cascades, and Why Gamma Rays Often Come Last

Physical Motivation

A nucleus does not necessarily have only one allowed decay path. The same parent state can decay to several daughter states, and a daughter state may itself be unstable. Nuclear data tables therefore show arrows with percentages: these are branching ratios.

If a parent can decay through two channels with partial decay constants λ_a and λ_b , the total decay constant is

$$\lambda = \lambda_a + \lambda_b. \quad (85)$$

The branching ratios are

$$b_a = \frac{\lambda_a}{\lambda}, \quad b_b = \frac{\lambda_b}{\lambda}, \quad b_a + b_b = 1. \quad (86)$$

Physical significance: The nucleus decays with the total rate λ , but once a decay occurs the probability of a particular channel is set by its partial rate divided by the total rate.

If the daughter is created in an excited state, it may emit one or more gamma photons while stepping down through the level scheme. Such a sequence is called a cascade. This is why gamma rays often appear as secondary radiation after a beta or alpha decay.

7.4 The Radioactive Decay Law

Physical Motivation

The defining assumption is that each nucleus has the same probability per unit time to decay, independent of how long it has already existed. This is a memoryless process. The macroscopic exponential law follows from that one microscopic statement.

Let $N(t)$ be the number of undecayed nuclei at time t . If each nucleus has probability λdt to decay during the short interval dt , then

$$\frac{dN}{dt} = -\lambda N. \quad (87)$$

Solving the differential equation gives

$$N(t) = N_0 e^{-\lambda t}. \quad (88)$$

Physical significance: The decay curve is exponential because the loss rate is proportional to how many unstable nuclei remain.

The activity is the number of decays per unit time:

$$\mathcal{A}(t) = \lambda N(t) = \lambda N_0 e^{-\lambda t}. \quad (89)$$

Activity has units of inverse time:

$$1 \text{ Bq} = 1 \text{ s}^{-1} = 1 \text{ decay/s}, \quad 1 \text{ Ci} = 3.7 \times 10^{10} \text{ Bq}. \quad (90)$$

Physical significance: Activity measures how many decays occur per second, not how much biological damage is caused. A high-activity source is not fully characterized until we also know the radiation type and energy.

7.4.1 Half-Life and Mean Lifetime

The half-life $T_{1/2}$ is defined by $N(T_{1/2}) = N_0/2$. Substituting Eq. (88),

$$\frac{1}{2} = e^{-\lambda T_{1/2}} \Rightarrow T_{1/2} = \frac{\ln 2}{\lambda}. \quad (91)$$

The mean lifetime τ is the average decay time of one nucleus:

$$\tau = \frac{\int_0^\infty t \lambda e^{-\lambda t} dt}{\int_0^\infty \lambda e^{-\lambda t} dt} = \frac{1}{\lambda}. \quad (92)$$

Thus

$$T_{1/2} = (\ln 2)\tau. \quad (93)$$

Physical significance: “Lifetime” is sometimes used loosely in speech. In formulas, $T_{1/2}$ and τ are different by the factor $\ln 2$.

7.5 How Decay Constants Are Measured

Physical Motivation

A lifetime measurement is not just a curve drawn through points. Radioactive decay is intrinsically statistical, so the scatter of the data is part of the physics. A good fit should not have every point exactly on the curve; it should have deviations consistent with the measurement uncertainties.

In a simple lifetime measurement, one produces a radioactive isotope, records the number of decays in time bins, and fits the counting rate to

$$\mathcal{A}(t) = \mathcal{A}_0 e^{-\lambda t} + B, \quad (94)$$

where B is a possible background rate. The quality of the fit is often checked with

$$\chi^2 = \sum_i \left(\frac{y_i - f(x_i)}{\Delta y_i} \right)^2. \quad (95)$$

If the uncertainties Δy_i are realistic and the model is correct, χ^2 should be of order the number of degrees of freedom $\nu = N_{\text{points}} - N_{\text{fit parameters}}$.

Physical significance: The usual correlation coefficient R^2 is not enough for a physics fit because it does not know the experimental uncertainties. A meaningful goodness-of-fit test must compare residuals to their errors.

7.6 Building Radioactivity: Activation

Physical Motivation

Radioactivity can be produced artificially. A stable material can be irradiated by particles or photons so that some nuclei are converted into radioactive isotopes. This is how many medical radioisotopes are produced in reactors or accelerators.

Suppose a radioactive isotope is produced at a constant rate R nuclei per second and decays with constant λ . Then

$$\frac{dN}{dt} = R - \lambda N, \quad N(0) = 0. \quad (96)$$

Solving,

$$N(t) = \frac{R}{\lambda} (1 - e^{-\lambda t}). \quad (97)$$

The activity during production is

$$\mathcal{A}(t) = \lambda N(t) = R (1 - e^{-\lambda t}). \quad (98)$$

Physical significance: Production saturates. After several lifetimes, the source is producing and losing nuclei at the same rate, so the activity approaches R . Irradiating much longer does not increase the activity substantially.

7.7 Decay Chains and Equilibrium

Physical Motivation

Often a radioactive parent decays into a daughter nucleus that is itself radioactive. Then the daughter abundance is controlled by two competing effects: it is produced by parent decay and destroyed by its own decay.

Consider a parent 1 with decay constant λ_1 , initially $N_1(0) = N_0$, decaying into a daughter 2 with decay constant λ_2 , initially $N_2(0) = 0$:

$$\frac{dN_1}{dt} = -\lambda_1 N_1, \quad (99)$$

$$\frac{dN_2}{dt} = \lambda_1 N_1 - \lambda_2 N_2. \quad (100)$$

The parent solution is

$$N_1(t) = N_0 e^{-\lambda_1 t}. \quad (101)$$

Substituting this into Eq. (100) gives

$$N_2(t) = \frac{\lambda_1 N_0}{\lambda_2 - \lambda_1} (e^{-\lambda_1 t} - e^{-\lambda_2 t}) \quad (\lambda_1 \neq \lambda_2). \quad (102)$$

The daughter activity is therefore

$$\mathcal{A}_2(t) = \lambda_2 N_2(t) = \frac{\lambda_1 \lambda_2 N_0}{\lambda_2 - \lambda_1} (e^{-\lambda_1 t} - e^{-\lambda_2 t}). \quad (103)$$

Long-lived parent, short-lived daughter. If $\lambda_1 \ll \lambda_2$, the parent changes very slowly compared with the daughter. After several daughter lifetimes,

$$\mathcal{A}_2(t) \approx \mathcal{A}_1(t). \quad (104)$$

Physical significance: This is secular equilibrium. The daughter decays as fast as it is produced, so its activity equals the parent activity even though the daughter may have a much shorter lifetime.

Comparable lifetimes. If $\lambda_1 < \lambda_2$ but not by a huge factor, the ratio $\mathcal{A}_2/\mathcal{A}_1$ approaches a constant after the daughter builds up:

$$\boxed{\frac{\mathcal{A}_2}{\mathcal{A}_1} \rightarrow \frac{\lambda_2}{\lambda_2 - \lambda_1}.} \quad (105)$$

Physical significance: This is transient equilibrium. Both activities eventually fall with the parent lifetime, but their ratio becomes nearly fixed.

Short-lived parent, long-lived daughter. If $\lambda_1 > \lambda_2$, the parent disappears quickly and leaves behind a daughter population that then decays on its own longer time scale:

$$\boxed{N_2(t) \approx N_0 e^{-\lambda_2 t} \quad \text{after the parent has decayed.}} \quad (106)$$

Physical significance: The source changes character: initially the parent dominates the radiation, but later the daughter controls the activity.

7.8 Radiation and Health

Physical Motivation

Activity tells us how many decays occur per second, but biological risk depends on what radiation reaches tissue, how much energy it deposits, and how damaging that radiation type is. We therefore need dose units, not only activity units.

Ionizing versus non-ionizing radiation. Ionizing radiation can remove electrons from atoms and break molecular bonds, including bonds in DNA. Non-ionizing radiation does not have a comparably clear ionization mechanism at ordinary exposure levels, although very intense non-ionizing radiation can still cause heating and other effects.

Radiation-protection principles. Two working principles are used in radiation protection:

- **ALARA:** exposures should be kept *As Low As Reasonably Achievable*. This is a practical rule, not a microscopic law.
- **LNT:** the *Linear No-Threshold* model assumes that risk is proportional to dose even at low doses, with no perfectly safe threshold.

Physical significance: LNT is a conservative regulatory model. It is used because high-dose harm is known and low-dose risk is hard to measure directly; it should not be confused with a precisely proven microscopic law at all dose levels.

7.9 Radiation Units

Physical Motivation

Different questions require different units. “How many nuclei decay per second?” is an activity question. “How much energy entered the material?” is an absorbed-dose question. “How risky is this for tissue?” is an equivalent dose question.

Exposure. Exposure measures ionization produced in air. The old unit is the roentgen:

$$\boxed{1 \text{ R} = 2.58 \times 10^{-4} \text{ C kg}^{-1} \quad \text{in air.}} \quad (107)$$

It is historically important, but less fundamental than absorbed dose.

Absorbed dose. Absorbed dose D measures deposited energy per unit mass:

$$\boxed{1 \text{ Gy} = 1 \text{ J kg}^{-1}, \quad 1 \text{ rad} = 10^{-2} \text{ Gy.}} \quad (108)$$

Physical significance: Absorbed dose is a physical energy-deposition quantity. It does not yet say whether the radiation was alpha, beta, gamma, neutrons, or protons.

Equivalent dose. Equivalent dose H weights the absorbed dose by a radiation quality factor Q :

$$\boxed{H = Q D, \quad 1 \text{ Sv} = 1 \text{ J kg}^{-1}, \quad 1 \text{ rem} = 10^{-2} \text{ Sv.}} \quad (109)$$

Typical quality factors are

$$\boxed{Q_\gamma \approx Q_\beta \approx Q_X \approx 1, \quad Q_p \approx 2, \quad Q_n \approx 5\text{--}20, \quad Q_\alpha \approx 20.} \quad (110)$$

Physical significance: Alpha particles are far more damaging per unit absorbed energy than gamma rays or beta particles because they deposit their energy densely over a short range. This is why an external alpha source may be easy to shield, while an internal alpha emitter can be biologically serious.

Key Takeaways

- Radioactive decay is probabilistic: the microscopic input is a decay probability per unit time, λ .
- Different decay modes are different ways to lower nuclear energy while preserving conservation laws.
- Activity is $\mathcal{A} = \lambda N$, measured in becquerel or curie; it is not itself a dose or a risk.
- Half-life and mean lifetime are related by $T_{1/2} = (\ln 2)\tau$.
- In decay chains, daughters may build up to secular or transient equilibrium with the parent.
- Radiation safety uses dose concepts: absorbed dose in gray and equivalent dose in sievert.

8 Lecture 06 — Radiation–Matter Interactions

Overview

To *detect* radiation we must first understand how it *interacts* with matter. Detection requires an interaction: this is why dark matter, which couples only gravitationally, cannot be seen by any conventional detector.

This lecture covers the three major classes of radiation: **heavy charged particles** (prototype: α), **light charged particles** ($\beta^\pm$), and **photons** (γ , X-rays). We also briefly discuss **neutrons**, which interact only via the nuclear force and present unique detection challenges. For each class we derive or motivate the key energy-loss mechanism and discuss the practical implications for shielding and detection.

8.1 Energy Scales and the Ionisation Picture

Physical Motivation

Before deriving any formula it is useful to set the stage with order-of-magnitude estimates. Nuclear radiation carries energies of order 10^2 keV–10 MeV, while chemical (atomic) bond energies are of order 1–30 eV. A single 5 MeV α particle therefore carries enough energy to ionise millions of atoms.

Why electrons dominate the energy loss. Consider an α particle of charge $ze = 2e$ passing through a material. The α feels both the nuclei and the orbital electrons. There are two key observations:

1. The α carries only a tiny fraction of the total particle number in the target: its effect on the *average* electric field in the material is negligible.
2. The nuclei are very heavy ($m_{\text{nuc}} \gg m_\alpha$) compared to electrons ($m_e \approx m_\alpha/8000$), but the *volume filling* of nuclei is negligible. Electrons, bound to atoms, fill essentially all of the volume.

The picture that emerges: the α traverses an extended *electron cloud*, losing energy in a large number of individually weak collisions. A single head-on collision with an electron transfers at most $\Delta E_{\text{max}} \approx 2.7$ keV (for a 5 MeV α), a fraction $\lesssim 0.1\%$ of the initial energy. The particle therefore travels *almost straight* through the medium — deflection is negligible.

8.2 Bethe-Bloch Formula for Heavy Charged Particles

Mathematical Setup

We derive the stopping power $S = -dE/dx$ (energy loss per unit path length) from first principles using the impulse approximation.

Single-collision momentum transfer. Consider an α of charge ze , mass M , and velocity v moving along the x -axis. An atomic electron sits at a perpendicular distance b (the *impact parameter*) from the trajectory. The α passes by the electron in a time $\Delta t \sim 2b/v$.

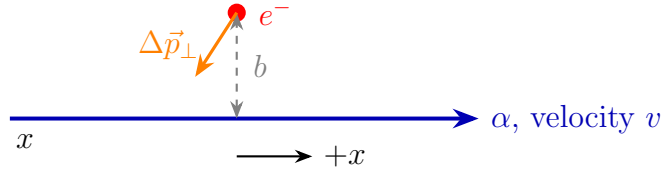


Figure 1: Geometry of a single α -electron collision. The α travels along the x -axis; the electron is at perpendicular distance b . By symmetry the parallel component of $\Delta\vec{p}$ averages to zero, leaving only a transverse kick.

By symmetry, the parallel component of the momentum transfer vanishes (the α is first attracted and then repelled symmetrically as it passes). Only the transverse component $\Delta p_{\perp}$ survives. Using the impulse approximation,

$$\Delta p_{\perp} = \int_{-\infty}^{\infty} F_{\perp} dt = \int_{-\infty}^{\infty} e E_{\perp} dt. \quad (111)$$

We change variables $dt = dx/v$ and recognise that the integral of the perpendicular electric field component over all x at fixed b can be evaluated with Gauss's law. Consider a cylindrical Gaussian surface of radius b and length L centred on the trajectory. For $L \rightarrow \infty$ the flux through the flat ends vanishes, and

$$\oint E_{\perp} dA = 2\pi b \int_{-\infty}^{\infty} E_{\perp} dx = \frac{ze}{\epsilon_0}. \quad (112)$$

Therefore

$$\int_{-\infty}^{\infty} E_{\perp} dx = \frac{ze}{2\pi\epsilon_0 b}. \quad (113)$$

Substituting into Eq. (111),

$$\Delta p_{\perp} = \frac{ze \cdot e}{2\pi\epsilon_0 bv} = \frac{ze^2}{2\pi\epsilon_0 bv}. \quad (114)$$

Significance: The transverse kick scales as $1/b$ and $1/v$ — a faster particle spends less time near each electron and receives a smaller kick.

Energy transfer to one electron. The electron recoil energy from a single collision at impact parameter b is

$$\Delta E(b) = \frac{(\Delta p_{\perp})^2}{2m_e} = \frac{z^2 e^4}{8\pi^2 \epsilon_0^2 m_e v^2 b^2}. \quad (115)$$

Significance: $\Delta E \propto z^2/v^2$ — heavier, slower, or more highly charged projectiles lose energy more rapidly.

Integration over all impact parameters

In an annular shell of radii $[b, b + db]$ and unit length along x , the number of electrons is $dn = n_e \cdot 2\pi b db$, where $n_e = N_0 \rho Z/A$ is the electron number density. Summing contributions from all impact parameters,

$$-\frac{dE}{dx} = \int_{b_{\min}}^{b_{\max}} \Delta E(b) n_e 2\pi b db = \frac{z^2 e^4 n_e}{4\pi \epsilon_0^2 m_e v^2} \int_{b_{\min}}^{b_{\max}} \frac{db}{b} = \frac{z^2 e^4 n_e}{4\pi \epsilon_0^2 m_e v^2} \ln\left(\frac{b_{\max}}{b_{\min}}\right). \quad (116)$$

The logarithm contains the physics of the cut-offs, which we now estimate.

Estimating $b_{\min}$. In a head-on collision the maximum energy transferred to an electron is $\Delta E_{\max} = 2m_e v^2$ (from non-relativistic elastic kinematics). Setting $\Delta E(b_{\min}) = \Delta E_{\max}$ gives

$$b_{\min} = \frac{ze^2}{4\pi\epsilon_0 m_e v^2}. \quad (117)$$

Estimating $b_{\max}$. The electron is bound to an atom with characteristic orbital frequency $\omega \sim I/\hbar$, where I is the mean ionisation energy of the target material. The interaction is effective only while the α is in the "vicinity" of the electron; more precisely, the interaction time $\Delta t \sim b/v$ must be shorter than the orbital period $2\pi/\omega$. This gives

$$b_{\max} \sim \frac{v}{\omega} \sim \frac{\hbar v}{I}. \quad (118)$$

Classical Bethe formula. Collecting terms,

$$\frac{b_{\max}}{b_{\min}} \propto \frac{2m_e v^2}{I}, \quad (119)$$

so

$$\boxed{-\frac{dE}{dx} \propto \frac{z^2 e^4 n_e}{m_e v^2} \ln\left(\frac{2m_e v^2}{I}\right)}. \quad (120)$$

Significance: $dE/dx \propto 1/v^2 \propto 1/E_k$ for non-relativistic projectiles — the particle loses energy *faster* as it slows down.

The Full Bethe-Bloch Formula

The complete quantum-relativistic treatment (Bethe 1930, Bloch 1933) gives:

Physical Law 8.1 Bethe-Bloch Formula

$$-\frac{dE}{dx} = \frac{z^2 e^4}{(4\pi\epsilon_0)^2} \frac{4\pi n_e}{m_e c^2 \beta^2} \left[\ln\left(\frac{2m_e c^2 \beta^2}{I(1-\beta^2)}\right) - \beta^2 \right], \quad (121)$$

where $\beta = v/c$, $n_e = N_0 \rho Z/A$ is the electron number density, z is the projectile charge number, and I is the mean ionisation energy of the target material.

Significance: The two new terms relative to our derivation are (1) the $-\beta^2$ inside the logarithm (relativistic correction — becomes important only near $v \approx c$) and (2) the replacement of $\epsilon_0 \rightarrow 1$ in Gaussian units. For α particles at ~ 5 MeV, rest energy ≈ 3.7 GeV, the kinetic energy is $\ll$ rest energy, so $\beta \ll 1$ and the formula reduces well to the classical limit.

Key dependences.

- $-dE/dx \propto z^2$: alpha particles ($z = 2$) lose energy $4\times$ faster than equally fast protons ($z = 1$).
- $-dE/dx \propto Z/A$ of the target material: higher atomic number means more electrons per unit mass, more stopping power.

- $-dE/dx \propto 1/v^2$: slower particles lose energy faster.
- $-dE/dx$ depends on the target only through n_e and I (not on nuclear structure): the formula is universal for all heavy charged particles.

8.3 Stopping Power, Bragg Peak, and Range

Physical Motivation

Because the stopping power increases as the particle slows ($-dE/dx \propto 1/E_k$), an α particle deposits very little energy near the entrance of a material and releases most of its energy just before stopping. This gives rise to the *Bragg peak*: a sharp maximum of energy deposition near the end of the range.

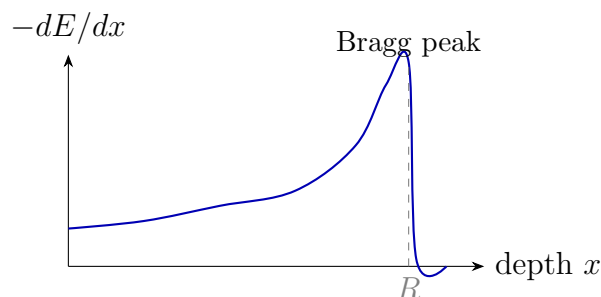


Figure 2: Schematic Bragg curve for a heavy charged particle. Energy deposition is low at the entrance and rises sharply near the end of range R (Bragg peak). A typical 5.5 MeV α in air has $R \approx 4$ cm; in solid tissue, $R < 1$ mm.

Physical Concept 8.1: Range of a Heavy Charged Particle

The range R is the total path length from the entrance to the point of rest:

$$R = \int_0^{E_0} \frac{dE}{(-dE/dx)}, \quad (122)$$

where the integral runs from zero kinetic energy back to the initial energy E_0 .

Significance: Unlike electrons (Section 8.4), heavy charged particles have a well-defined range with only small straggling. This property underpins proton/ion therapy in cancer treatment (Bragg peak can be placed at the tumour depth).

Note:

A practical rule: α particles from radioactive decay (~ 5 MeV) travel ~ 4 cm in air but < 0.1 mm in tissue or aluminium foil. A single sheet of paper stops them.

Shielding implications. From Eq. (120), $-dE/dx \propto Z^2/A$ of the target. To stop α particles efficiently one wants a high- Z , high-density material (many electrons per unit volume), but in practice even low- Z materials work because the range is intrinsically short.

8.4 Beta Particles in Matter

Physical Motivation

An electron or positron ($\beta^\pm$) has mass $m_e \approx m_\alpha/8000$. In a head-on collision with an atomic electron, the projectile can transfer virtually all of its kinetic energy to the target (identical particles). The basic assumption of the Bethe-Bloch treatment — that the projectile travels straight, losing negligible fractional energy per collision — breaks down completely.

Chaotic motion. Each collision can deflect the β particle significantly. The trajectory is a random walk, and there is *no well-defined range*: different electrons with the same initial energy stop at different depths. One cannot speak of a “range” as one does for α particles.

Bremsstrahlung. A second loss channel, absent for heavy projectiles, is important for electrons: whenever a fast electron is decelerated by the nuclear Coulomb field, it emits a photon (*Bremsstrahlung*, German: “braking radiation”). The total stopping power for electrons is:

$$\boxed{-\frac{dE}{dx} = \left(-\frac{dE}{dx}\right)_{\text{coll}} + \left(-\frac{dE}{dx}\right)_{\text{rad}}}. \quad (123)$$

The radiative component scales as $Z^2 E_k$ and becomes dominant at high energies or in high- Z materials.

Consequence 8.2 Beta Shielding Paradox

Shielding β radiation with a high- Z material is *counterproductive*: the electrons stop quickly by the Bethe-Bloch mechanism, but the Bremsstrahlung photons produced escape the shield and may be more penetrating and biologically damaging than the original electrons. The correct choice is a low- Z absorber (e.g., plastic, water) followed, if needed, by a high- Z layer to absorb the resulting X-rays.

8.5 Photon Interactions with Matter

Conceptual Background

Photons carry no electric charge, so they do *not* experience the continuous Coulomb energy loss that characterises charged particles. Instead, a photon either traverses the material without interaction or is removed from the beam in a single event. This binary behaviour leads to exponential attenuation rather than a definite range.

There are three dominant processes, depending on photon energy:

1. **Photoelectric effect** — dominant at low E_γ
2. **Compton scattering** — dominant at intermediate E_γ
3. **Pair production** — dominant at high E_γ

8.5.1 Photoelectric Effect

Mathematical Setup

The photon is completely absorbed by an atomic electron, which is ejected with kinetic energy

$$T_e = E_\gamma - E_b, \quad (124)$$

where E_b is the binding energy of the ejected electron (typically $\ll E_\gamma$ for nuclear radiation). The electron carries a *sharp, discrete* energy equal to $E_\gamma - E_b$.

Significance: A photopeak at energy $E_\gamma - E_b \approx E_\gamma$ in a detector spectrum is the signature of photoelectric absorption. It is the most useful feature for γ -ray spectroscopy.

8.5.2 Compton Scattering

Mathematical Setup

The photon scatters off a (quasi-free) electron, transferring part of its energy. Conservation of 4-momentum gives the Compton formula for the scattered photon energy:

$$E_{\gamma'} = \frac{E_\gamma}{1 + \frac{E_\gamma}{m_e c^2} (1 - \cos \theta)}, \quad (125)$$

where θ is the photon scattering angle. The *recoil electron* receives $T_e = E_\gamma - E_{\gamma'}$.

Compton edge. The maximum electron recoil energy occurs at $\theta = \pi$ (backscatter):

$$T_{e,\max} = E_\gamma - \frac{E_\gamma}{1 + 2E_\gamma/m_e c^2}. \quad (126)$$

This creates the characteristic *Compton edge* in the detector spectrum: a sharp drop at $T_{e,\max}$, with a continuum of electron energies below it.

Significance: Unlike photoelectric absorption, a single Compton scatter does not remove the photon — a degraded photon exits the interaction and may scatter again or escape. The electron recoil forms a *continuous spectrum* (not a sharp line), making Compton events harder to use for spectroscopy.

8.5.3 Pair Production

Physical Motivation

Above the threshold $E_\gamma = 2m_e c^2$, the photon can materialise into an electron–positron pair in the Coulomb field of a nucleus:

$$\gamma + \text{nucleus} \rightarrow e^- + e^+ + \text{nucleus}, \quad E_{\gamma,\text{thr}} = 2m_e c^2 = 1.022 \text{ MeV}. \quad (127)$$

The 511 keV annihilation photons from $e^+ + e^- \rightarrow 2\gamma$ are a distinctive signature.

Escape peaks. If one or both annihilation photons escape the detector,

$$\text{single escape peak: } E_{\text{dep}} = E_{\gamma} - 511 \text{ keV}, \quad (128)$$

$$\text{double escape peak: } E_{\text{dep}} = E_{\gamma} - 1022 \text{ keV}. \quad (129)$$

Significance: The appearance of two lines at $E_{\gamma} - 511 \text{ keV}$ and $E_{\gamma} - 1022 \text{ keV}$ is an unambiguous signature of pair production and allows identification of $E_{\gamma} > 1.022 \text{ MeV}$ sources.

8.5.4 Attenuation Coefficient and Dominance Regions

Mathematical Setup

Since a photon is either removed or not, the beam obeys exponential attenuation:

$$I(x) = I_0 e^{-\mu x}, \quad (130)$$

where the total linear attenuation coefficient μ is the sum of partial coefficients:

$$\mu = \mu_{\text{photo}} + \mu_{\text{Compton}} + \mu_{\text{pair}}. \quad (131)$$

This is measured by placing absorbers of varying thickness x in the beam and fitting the exponential decay.

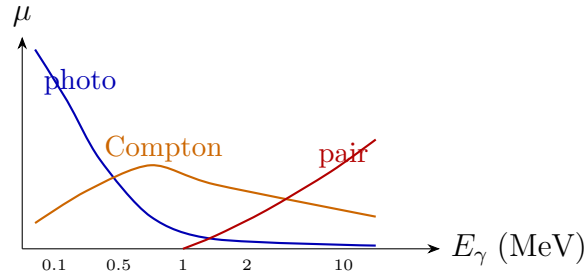


Figure 3: Schematic energy dependence of the three photon attenuation coefficients in a medium- Z material. Photoelectric dominates at low energy, Compton at intermediate energies, and pair production above $\sim 1 \text{ MeV}$.

8.6 Neutrons in Matter

Physical Motivation

Neutrons carry no electric charge and do not interact electromagnetically. They interact only via the strong nuclear force — directly with nuclei, not with the electron cloud. Since nuclei fill a tiny fraction of the volume of matter, neutrons pass through most materials almost unimpeded. This makes them both very difficult to detect and very difficult to shield against.

Elastic scattering. The dominant energy-loss mechanism for neutrons is elastic nuclear scattering:



By the kinematics of elastic scattering, the maximum energy transfer is

$$\Delta E_{\max} = \frac{4m_n M_A}{(m_n + M_A)^2} E_n. \quad (133)$$

This is maximised when $M_A = m_n$: scattering off *hydrogen nuclei (protons)* is the most efficient moderator.

Claim 8.1 Water as Neutron Moderator

Hydrogen ($M_A = m_n$) allows $\Delta E_{\max} = E_n$ per collision — all the energy can be transferred. This is why water, polyethylene, and concrete (hydrogen-rich materials) are excellent neutron moderators and shields. High- Z materials (lead, iron) are nearly useless for neutron slowing because $m_n \ll M_A$.

Attenuation law for neutrons. Like photons (binary interaction), neutrons obey

$$I(x) = I_0 e^{-n\sigma x}, \quad (134)$$

where σ is the total nuclear scattering cross section and n is the number density of target nuclei.

Detection strategy: conversion. Because neutrons cannot be detected directly (they deposit no ionisation energy in most detectors), the standard strategy is *conversion*: induce a nuclear reaction that produces a charged particle, then detect the charged particle:



The resulting α particle is easily detected. Other converter reactions use ${}^3\text{He}$, ${}^6\text{Li}$, and ${}^{235}\text{U}$ (fission chambers).

Significance: Neutron detection is fundamentally harder than photon or charged-particle detection: there is no material in which a neutron simply stops and deposits energy. Converter-based detection is inherently indirect and introduces a factor of uncertainty.

8.7 Cloud Chamber — A Direct Window into Radiation

Physical Motivation

The cloud chamber (Wilson, 1911) is the simplest radiation detector: it makes individual particle tracks directly visible. Alcohol vapour is maintained in a supersaturated state by cooling (using dry ice). An ionising particle passing through triggers droplet nucleation along its track — the track becomes visible as a line of droplets.

- α tracks: short, thick, straight (high dE/dx , small range, negligible deflection).
- β tracks: long, thin, tortuous (lower dE/dx , significant scattering at each collision).

- Applying a magnetic field: $\vec{F} = q\vec{v} \times \vec{B}$ curves the tracks, allowing charge-sign and momentum determination.
- Limitations: slow reset time ($\gg 1$ ms), unsuitable for high-rate measurements; replaced by wire chambers, silicon detectors, etc. in modern experiments.

Key Takeaways

- **Heavy charged particles (α , p , ions):** energy loss described by Bethe-Bloch, $-dE/dx \propto z^2 Z/v^2$; well-defined range; Bragg peak at end of range; easily shielded.
- **Beta particles ($e^\pm$):** no well-defined range (large deflections); Bremsstrahlung adds a radiative loss channel; shield with low- Z material.
- **Photons:** three discrete interaction mechanisms; exponential attenuation $I = I_0 e^{-\mu x}$; photoelectric dominant at low E_γ , Compton intermediate, pair production above 1.022 MeV.
- **Neutrons:** no EM interaction; attenuate exponentially like photons; hydrogen most effective moderator; detected via conversion reactions.
- The *Bragg peak* and the *Compton continuum* are experimentally central signatures that recur throughout detector physics and spectroscopy.

9 Lecture 07 — Nuclear Detectors: Resolution, Efficiency, Tracking

Overview

This lecture begins with a correction to the Bethe-Bloch formula from Lecture 6, clarifies the mass-dependence of stopping power, and then develops the remaining detector topics: the statistical origin of energy resolution, detector efficiency, particle tracking, particle identification, time-of-flight for neutrons, and a look at real experimental spectra.

9.1 Correction to Lecture 6: Mass Dependence in Bethe-Bloch

Physical Motivation

In Lecture 6 we quoted the Bethe-Bloch formula and stated that the projectile mass does *not* appear. A question from a student prompted a more careful discussion: where does the projectile mass enter, and why can the range of an electron be larger than that of a proton at the same kinetic energy?

Why the projectile mass is absent from $-dE/dx$. Recall how we derived the momentum transfer to a single electron at impact parameter b :

$$\Delta p_{\perp} = \frac{ze^2}{2\pi\epsilon_0 b v}. \quad (136)$$

The energy transferred to that electron is

$$\Delta E(b) = \frac{(\Delta p_{\perp})^2}{2m_e} = \frac{z^2 e^4}{8\pi^2 \epsilon_0^2 m_e v^2 b^2}. \quad (137)$$

The *electron* mass m_e appears — not the projectile mass M . This is because the entire derivation was formulated in terms of *velocity*: the projectile's mass never entered the impulse integral. The factor $1/(2m_e)$ is the electron's kinetic energy for a given momentum kick.

Physical Concept 9.1: W-Value (Energy per Ion Pair)

The W -value of a material is the average energy deposited per electron-ion pair created:

$$W = \frac{E_{\text{deposited}}}{N_{\text{ion pairs}}}. \quad (138)$$

Typical values: $W_{\text{gas}} \approx 30 \text{ eV}$, $W_{\text{silicon}} \approx 3 \text{ eV}$.

Where the projectile mass does appear. The formula $-dE/dx \propto z^2/v^2$ is a function of *velocity*. When we express it as a function of kinetic energy $E_k = \frac{1}{2}Mv^2$ (non-relativistically), we get $v^2 = 2E_k/M$, so

$$-\frac{dE}{dx} \propto \frac{z^2 M}{E_k}. \quad (139)$$

The projectile mass enters *only through the velocity-energy conversion*. At the *same velocity* the formula is independent of M ; at the *same kinetic energy* it scales as $z^2 M$.

Comparing α and proton at the same kinetic energy. Take $E_\alpha = E_p$. Since $m_\alpha = 4m_p$,

$$\frac{1}{2}m_\alpha v_\alpha^2 = \frac{1}{2}m_p v_p^2 \Rightarrow \frac{v_\alpha^2}{v_p^2} = \frac{m_p}{m_\alpha} = \frac{1}{4}. \quad (140)$$

With $z_\alpha = 2$ and $z_p = 1$,

$$\frac{(-dE/dx)_\alpha}{(-dE/dx)_p} = \frac{z_\alpha^2}{z_p^2} \cdot \frac{v_p^2}{v_\alpha^2} = 4 \times 4 = \boxed{16}. \quad (141)$$

Significance: An α particle loses energy *sixteen times faster* than a proton of the same kinetic energy. Heavier and more highly-charged projectiles have shorter ranges.

Why electrons have larger ranges than protons at the same energy. An electron at energy E_k has $v^2 = 2E_k/m_e$, which is $\approx 1836\times$ larger than a proton's v^2 at the same energy. Since $-dE/dx \propto 1/v^2$, the stopping power on the electron is $\sim 1836\times$ smaller. (At typical decay energies electrons are already ultra-relativistic, making the contrast even sharper.)

9.2 Detector Types — Comparison

Conceptual Background

Three detector families were introduced at the end of Lecture 6. Here we compare them systematically across the properties that matter experimentally: energy resolution, timing resolution, cost, and practical constraints.

Property	Scintillator	Gas detector	Solid-state
Energy resolution	moderate ($\sim$ few %)	poor–moderate ($\sim$ few %)	excellent (~ 0.1 %)
Timing resolution	excellent ($\sim$ ps–ns)	poor ($\sim \mu$ s–ms)	good ($\sim$ ns)
Cost per area	low	moderate	very high
Scalable to large size	yes (cheap plastic)	yes (gas)	no
Works as reaction target	no	yes	no

Table 1: Comparison of the three main detector families.

Why gas detectors are slow. Drift velocity of electrons in gas is $\sim 1\text{ cm}/\mu\text{s}$, so a detector of size $\sim \text{cm}$ takes $\sim \mu\text{s}$ to collect the signal. This limits the maximum counting rate to $\sim 10\text{--}100\text{ kHz}$.

Advantages of gas detectors despite low speed. Gas can *surround* a nuclear reaction, yielding 4π solid-angle coverage and near-unity geometric efficiency. Reactions can occur *inside* the detector volume. This is the main reason gas detectors remain in use despite their slow response.

9.3 Statistical Origin of Energy Resolution

Mathematical Setup

All three detector types ultimately convert deposited energy into a number of charge carriers (electrons or photon-electron pairs). The statistical fluctuation in this number sets a fundamental lower bound on energy resolution.

Number of charge carriers. A particle depositing energy E produces, on average,

$$\bar{N} = \frac{E}{W}, \quad (142)$$

where W is the W -value. For Poisson statistics (independent events),

$$\sigma_N \approx \sqrt{\bar{N}}. \quad (143)$$

Relative energy resolution. Since $E \propto N$, the relative energy resolution is

$$\frac{\sigma_E}{E} = \frac{\sigma_N}{\bar{N}} = \frac{1}{\sqrt{\bar{N}}} = \sqrt{\frac{W}{E}}. \quad (144)$$

This is the *ideal* (quantum-limited) resolution.

Worked Example 9.1 (Resolution for a 13 MeV Alpha Particle)

For $E_\alpha = 13 \text{ MeV}$:

Gas detector ($W \approx 30 \text{ eV}$):

$$\bar{N}_{\text{gas}} \approx \frac{13 \times 10^6}{30} \approx 4 \times 10^5, \quad \frac{\sigma_E}{E} \approx \frac{1}{\sqrt{4 \times 10^5}} \approx 0.5 \%.$$

Solid-state ($W \approx 3 \text{ eV}$):

$$\bar{N}_{\text{solid}} \approx \frac{13 \times 10^6}{3} \approx 4 \times 10^6, \quad \frac{\sigma_E}{E} \approx \frac{1}{\sqrt{4 \times 10^6}} \approx 0.1 \%.$$

The factor ~ 3 improvement in resolution comes from the factor ~ 10 larger number of charge carriers, i.e., $\sqrt{10} \approx 3.16$.

Significance: In practice, gas detectors achieve resolutions significantly worse than this ideal (electronic noise, recombination, gain non-uniformity), while solid-state detectors routinely approach the ideal $\sim 0.1 \%$, making them the instrument of choice for precision spectroscopy.

Note:

For scintillators the analogous quantity is the light yield: the number of optical photons produced per MeV. The photon-to-electron conversion in the PMT introduces additional fluctuations; typical energy resolution is a few percent.

9.4 Detector Efficiency

Physical Motivation

A detector is only useful if it actually registers the particles we want to measure. The *efficiency* tells us what fraction of emitted particles ends up contributing to our data. For rare processes — neutrinos, dark matter candidates, exotic nuclear reactions — maximising efficiency is as important as any other design consideration.

Physical Concept 9.2: Detector Efficiency

$$\varepsilon \equiv \frac{N_{\text{detected}}}{N_{\text{emitted}}}. \quad (145)$$

This quantity depends on the specific definition of “emitted,” which must always be stated clearly when quoting an efficiency.

Two components of efficiency. Geometric acceptance ($\Omega/4\pi$): A point source emitting isotropically at distance d from a detector of area A subtends a solid angle $\Omega \approx A/d^2$ (for $d \gg \sqrt{A}$). The fraction of particles that reach the detector is

$$\varepsilon_{\text{geo}} = \frac{\Omega}{4\pi} \approx \frac{A}{4\pi d^2}. \quad (146)$$

This scales as $1/d^2$ — moving the detector twice as far away reduces geometric efficiency by a factor of 4.

Intrinsic efficiency: Of the particles that enter the detector, what fraction actually produce a signal?

- Charged particles traversing an active volume: $\varepsilon_{\text{int}} \approx 1$ (every ionising particle deposits energy).
- Photons: binary interaction, exponential probability. A photon of attenuation coefficient μ in a detector of thickness x has

$$\varepsilon_{\text{int}} = 1 - e^{-\mu x} \approx \mu x \quad (\mu x \ll 1). \quad (147)$$

To detect photons efficiently, one wants large μ (high- Z material) and large x , but larger x costs spatial resolution and money.

Total efficiency and its importance.

$$\varepsilon_{\text{tot}} = \varepsilon_{\text{geo}} \times \varepsilon_{\text{int}}. \quad (148)$$

The statistical uncertainty on a rate measurement is $\propto 1/\sqrt{N_{\text{detected}}} \propto 1/\sqrt{\varepsilon_{\text{tot}}}$. Doubling efficiency is equivalent to quadrupling measurement time.

Note:

DUNE (neutrino experiment): 6 m × 14 m × 62 m detector volume containing 17 000 tonnes of liquid argon at cryogenic temperatures. The extreme scale is necessary because the cross section for neutrino-nucleus interactions is $\sim 10^{-38} \text{ cm}^2$, requiring enormous target mass to achieve measurable statistics.

Strategies for maximising efficiency.

1. Enlarge detector area or use an array of detectors.
2. Move detector closer to source (geometric efficiency $\propto 1/d^2$).
3. Surround the reaction (gas detectors can provide 4π coverage).
4. Improve data acquisition: fast electronics can reduce dead time and allow a higher fraction of events to be recorded.
5. Improve event selection: using trigger criteria to reduce background can effectively raise the signal fraction even if the raw geometric efficiency stays the same.

9.5 Particle Tracking

Physical Motivation

Energy and timing alone often do not tell us everything. Knowing the *trajectory* of a particle provides: its direction of emission ($\rightarrow$ reaction kinematics), whether it scattered or decayed in flight ($\rightarrow$ topology), and its curvature in a magnetic field ($\rightarrow$ momentum and charge sign).

General principle. A tracker measures the position of a particle at several known planes or volumes. By fitting a curve to these space points, one reconstructs the 3D trajectory.

Multi-Wire Proportional Chamber (MWPC)

A MWPC is a gas detector in which the usual single anode wire is replaced by an array of many parallel wires with pitch $\sim 1\text{--}2\text{ mm}$. Each wire is read out independently.

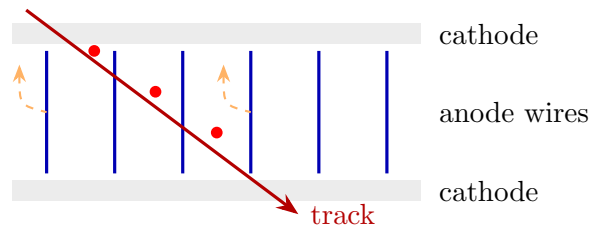


Figure 4: Schematic MWPC cross-section. A charged particle crossing the active volume produces ion clusters (red dots). Electrons drift to the nearest anode wire and trigger avalanche multiplication. The wire that fires identifies the x -coordinate of the hit with $\sim\text{mm}$ precision. Two chambers oriented perpendicularly provide 2D (and, combining with drift time, 3D) information.

How the MWPC gives position. Near each wire the electric field $\propto 1/r$ causes avalanche multiplication close to the wire surface. The avalanche signal is confined to the wire that was closest to the ionisation cluster. Reading out each wire separately maps the *hit position* to a single wire channel — i.e., to a known x -coordinate.

A second MWPC with wires rotated by 90° provides y -coordinates. Combining both planes gives a 2D hit position; adding the drift time in the gas gives a third coordinate.

Limitations and successors. MWPC wire pitch is limited to ~ 1 mm; stringing thousands of wires is labour-intensive. Modern replacements are *Micro-Pattern Gas Detectors* (MPGD):

- **Micromegas:** fine metallic mesh $\sim 100\ \mu\text{m}$ above a patterned electrode; strong field below mesh produces avalanche.
- **GEM (Gas Electron Multiplier):** insulating foil with micro-holes; each hole acts as an independent avalanche region.

Both achieve spatial resolution $\sim 50\text{--}100\ \mu\text{m}$ and can be fabricated by photolithographic techniques.

Silicon Strip and Pixel Detectors

A thin ($\lesssim 1$ mm) silicon diode is segmented into narrow strips (pitch $\sim 50\text{--}100\ \mu\text{m}$) or pixels. Each strip/pixel is read out independently. A charged particle crossing the detector creates electron-hole pairs; the strip closest to the track fires.

By stacking several layers (each with orthogonal strip orientation) one reconstructs a full 3D track, as illustrated in Fig. 5.

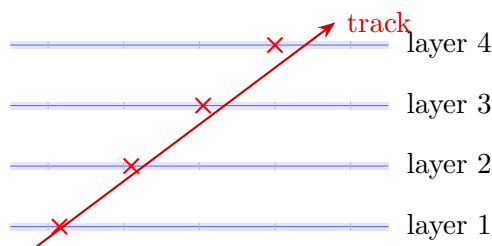


Figure 5: Four silicon strip detector layers. Each layer records a hit position (cross marks). Connecting the hits reconstructs the 3D trajectory. Alternating strip orientations (0° , 90° , ...) in successive layers provide both transverse coordinates.

Note:

This is the principle of the central tracking detectors in ATLAS and CMS at the LHC. The “giant wheel” surrounding the beam axis is largely a silicon pixel tracker plus outer strip layers, whose primary purpose is to reconstruct particle trajectories and measure their momenta via curvature in the axial magnetic field.

9.6 Particle Identification

Physical Motivation

So far detectors measured only energy; they were agnostic to particle species. In nuclear physics experiments one often needs to know *which particle* produced the signal (proton, deuteron, α , heavier fragment?). Several techniques exploit the mass and charge dependence of energy loss.

Method 1: dE/dx –Range

From the Bethe-Bloch formula, at the same energy E_k , $-dE/dx \propto z^2 M/E_k$ and the range $R \propto E_k/(z^2 M)$. A gas (or silicon) detector that is thick enough to stop the particle records both the total deposited energy *and* the path length (from the track or the stopping position). The combination $(dE/dx, \text{range})$ uniquely identifies (z, M) .

Method 2: ΔE – E Telescope

Physical Concept 9.3: ΔE – E Telescope

A ΔE – E telescope consists of two detectors in series:

1. A **thin** (ΔE) detector: the particle passes through, depositing a small fraction of its energy.
2. A **thick** (E) detector: the particle stops here, depositing its remaining energy.

One records the pair $(\Delta E, E_{\text{stop}})$ for each event.

Why does this identify the particle? For a non-relativistic particle of charge z , mass M , and kinetic energy E_k ,

$$-\frac{dE}{dx} \propto \frac{z^2 M}{E_k}. \quad (149)$$

The energy lost in the ΔE layer is approximately $\Delta E \approx (dE/dx) \cdot \Delta x \propto z^2 M/E_k$. But $E_k = \Delta E + E_{\text{stop}}$, so

$$\Delta E \cdot (\Delta E + E_{\text{stop}}) \propto z^2 M. \quad (150)$$

Plotting ΔE vs. E_{stop} , each species (z, M) traces out a **characteristic hyperbola-like curve**. Different isotopes and elements separate cleanly on this plot.

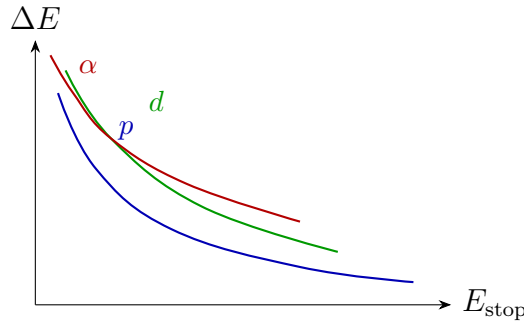


Figure 6: Schematic ΔE – E plot. Each particle species traces a distinct locus. Protons ($z=1, M=1$), deuterons ($z=1, M=2$), and α particles ($z=2, M=4$) separate cleanly, allowing unambiguous identification.

Significance: The ΔE – E telescope is one of the most widely used particle identification methods in nuclear physics. It requires only two detectors and is straightforward to implement with either scintillators or solid-state devices.

Method 3: Gamma–Charged Coincidence

The γ -ray spectrum emitted by a nucleus is a unique fingerprint of that isotope (analogous to atomic emission lines). If a nuclear reaction produces a specific isotope in an excited state, the characteristic de-excitation γ rays can be measured in coincidence with the charged particle. Together, $(E_{\text{charged}}, E_{\gamma})$ identify both the isotope and its excitation energy.

9.7 Time-of-Flight for Neutron Energy Measurement

Physical Motivation

Neutrons have no charge and cannot be detected directly. The standard approach is *conversion*: a nuclear reaction produces a charged particle that can be detected. However, for ${}^6\text{Li}$ -based converters (reaction $n + {}^6\text{Li} \rightarrow {}^4\text{He} + {}^3\text{H}$), the light output is dominated by the Q -value of the conversion reaction ($Q = 4.78\text{ MeV}$), *not* by the incoming neutron energy. There is therefore no direct energy-to-light correlation, and one cannot measure E_n from pulse height alone.

Physical Concept 9.4: Time-of-Flight (ToF) Method

Produce neutrons in **pulses** at a known position. Start a clock when the pulse is produced (e.g., using the beam pick-up signal from the accelerator); stop the clock when a neutron is detected at distance d . The neutron energy is then:

$$E_n = \frac{1}{2}m_n \left(\frac{d}{t_{\text{ToF}}} \right)^2, \quad (151)$$

valid for $E_n \ll m_n c^2 = 939.6\text{ MeV}$ (non-relativistic regime).

Significance: ToF is the standard technique for measuring neutron energy spectra in nuclear physics. The energy resolution improves with longer flight paths (better time-to-energy discrimination) but at the cost of reduced geometric efficiency.

Experimental example. In a recent experiment at a small accelerator in Braunschweig (Germany), protons impinged on an ${}^{18}\text{O}$ target:



The neutron energy spectrum was measured by ToF using a ${}^6\text{LiF}$ scintillator detector. The proton beam was pulsed; the beam pick-up signal started the clock, and the scintillator signal stopped it. From Eq. (151), each recorded time difference converts directly to a neutron energy.

9.8 Experimental Spectra

Alpha Spectrum of the ${}^{228}\text{Th}$ Decay Chain

Physical Motivation

Thorium-228 undergoes a long series of successive α and β decays. Each α -decaying isotope in the chain emits α particles at a *unique, discrete* energy — a direct consequence

of energy-momentum conservation and the discrete nuclear energy levels.

The chain $^{228}\text{Th} \rightarrow \dots \rightarrow ^{208}\text{Pb}$ includes:



A silicon detector placed near a ^{228}Th source records all α peaks simultaneously, since the chain quickly reaches secular equilibrium.

Key features of the measured spectrum:

- **Sharp peaks:** each peak corresponds to one α -emitting isotope; peak width reflects the detector energy resolution ($\sim 0.1\%$ for silicon at room temperature).
- **No background:** charged particles deposit *all* their energy in the detector; there is no Compton continuum analogue.
- **Low-energy tail:** “incomplete energy collection” — some α -produced ions recombine near the surface or in the dead layer before being swept out, leading to a slight tail below the main peak. This is a detector artefact, not physics.
- **Axis in channels:** the ADC converts the signal height (voltage) to a channel number. Calibration maps channel $\leftrightarrow$ energy using the known α energies of the isotopes in the chain.

Gamma Spectrum of ^{60}Co with HPGe Detector

^{60}Co decays via β^- to an excited state of ^{60}Ni , which de-excites by emitting two γ rays in cascade: $E_{\gamma_1} \approx 1173\text{ keV}$ and $E_{\gamma_2} \approx 1333\text{ keV}$.

An HPGe (high-purity germanium) detector provides energy resolution of $\sim 0.1\%$, resolving these two lines clearly. The spectrum shows (on a log scale):

Photo peaks. Two narrow peaks at 1173 and 1333 keV. A photon is completely absorbed by the photoelectric effect (or a sequence of interactions whose net result is complete energy deposition); all its energy goes to one electron (minus the small binding energy). The resulting sharp spike is the *full-energy peak* (commonly called the *photo peak*).

Compton continuum. The dominant feature across most of the spectrum. A γ scatters off a quasi-free electron (Compton scattering), depositing a *continuous* range of electron energies from near zero up to the Compton edge:

$$T_{e,\text{max}} = E_\gamma - \frac{E_\gamma}{1 + 2E_\gamma/m_e c^2}. \quad (153)$$

For $E_{\gamma_1} = 1173\text{ keV}$, this gives $T_{e,\text{max}} \approx 963\text{ keV}$; for $E_{\gamma_2} = 1333\text{ keV}$, $T_{e,\text{max}} \approx 1118\text{ keV}$. The spectrum shows **two Compton edges** — characteristic step-like drops at these energies — visible even on the log scale.

Log scale is essential. On a linear scale only the two photo peaks are visible; all other features are invisible. The log scale reveals:

1. Two Compton edges (as sharp drops in the continuum).
2. The region between the Compton edge and the photo peak (“Compton valley”) contains contributions from multiple scatters.
3. A 511 keV annihilation peak (weak, from pair production followed by positron annihilation) is visible in principle but requires a strong source.

Note:

The photo peak is not *exclusively* from the photoelectric effect. Compton scattering followed by complete absorption of the scattered photon also deposits the full photon energy. Monte Carlo simulations show that pair production (above threshold) and multiple Compton scatters all contribute to the full-energy peak.

Key Takeaways

- **Bethe-Bloch correction:** the projectile mass M does not appear in $-dE/dx(v)$; it enters only when converting to $-dE/dx(E_k)$ via $v^2 = 2E_k/M$. At the same energy, stopping power ratios go as z^2M . An α is stopped $16\times$ more efficiently than a proton at the same E_k .
- **Energy resolution:** limited by $\sigma_E/E = \sqrt{W/E}$. Silicon ($W \approx 3\text{ eV}$) gives $\sim 0.1\%$; gas ($W \approx 30\text{ eV}$) gives ideal $\sim 0.5\%$ but worse in practice.
- **Efficiency:** $\varepsilon_{\text{tot}} = \varepsilon_{\text{geo}} \times \varepsilon_{\text{int}}$; geometric efficiency $\propto 1/d^2$. For rare processes, maximising efficiency is a primary design driver.
- **Tracking:** MWPC, MPGD (micromegas/GEM), silicon strip/pixel detectors all reconstruct 3D particle trajectories; silicon gives $\sim 50\text{ }\mu\text{m}$ resolution.
- **Particle ID:** ΔE – E telescope identifies species via $\Delta E \cdot E_{\text{tot}} \propto z^2M$; each species maps to a distinct curve in the $(\Delta E, E_{\text{stop}})$ plane.
- **Neutron energy:** measured by ToF since direct energy-to-light conversion fails in conversion detectors. $E_n = \frac{1}{2}m_n(d/t)^2$ (non-relativistic).

10 Lecture 08 — Theory of Radioactive Decay & Alpha Decay

Lecture date: 2 June 2026 — *Lecturer: Dr. Moshe Friedman.*

Overview

This lecture opens the chapter on *radioactive decay* (Krane Ch. 7). It splits cleanly into two halves. The first half is a *general, model-independent* theory of decay: why a decaying state cannot be a static quantum superposition, why we must split the Hamiltonian into a stable part plus a small perturbation, how Fermi's golden rule then produces the decay constant, and how the finite lifetime of any decaying state forces it to acquire a *Lorentzian energy width* $\Gamma = \hbar/\tau$. The second half specialises to *alpha decay*: the Q -value and decay kinematics, the empirical Geiger–Nuttall law, the Gamow tunnelling theory that explains it, the role of angular momentum, and the selection rules.

10.1 A Decaying State Is Not a Static Superposition

Conceptual Background

Every decay we will study — β^- ($n \rightarrow p + e^- + \bar{\nu}_e$), β^+ ($p \rightarrow n + e^+ + \nu_e$), α emission, and γ de-excitation — is fundamentally a transition from one quantum state to another: an initial nuclear state ψ_i evolves into a final state ψ_f , possibly with extra particles created in the process. The natural first guess, drawing on ordinary quantum mechanics, is to write the system as a fixed superposition

$$|\psi\rangle = A |\psi_a\rangle + B |\psi_b\rangle. \quad (154)$$

Why this fails. A *stationary* superposition like (154) does *not* decay. Recall the deuteron (Lecture 4): it is a fixed admixture of S - and D -waves, $\psi_d = A\psi_S + B\psi_D$. It is *always* simultaneously S and D ; measuring it now or in a year gives the *same* probabilities $|A|^2$, $|B|^2$. The mixing coefficients are time-independent. A photon prepared as a superposition of two polarisations stays in that superposition no matter how far it travels — it does not spontaneously “decay” into one component. In short, (154) describes a static object, whereas radioactive decay is intrinsically a *time-dependent* process. We therefore need a different description.

Mathematical Setup — Split the Hamiltonian

The resolution is to write the full Hamiltonian as an unperturbed piece plus a small perturbation:

$$H = \underbrace{H_0}_{\text{stable configuration}} + \underbrace{V'}_{\text{decay interaction}}, \quad V' \ll H_0. \quad (155)$$

Significance: H_0 describes the relatively stable nucleus whose eigenstates are the levels we actually measure (energies, spins, parities); the small term V' is the interaction physically responsible for the decay — electromagnetic for γ , the weak interaction for β , the strong (plus Coulomb) for α .

Why V' is genuinely small. The justification is a separation of time scales. A nucleus exists in a well-defined state for a characteristic time set by its lifetime — which can be seconds, or for ^{235}U hundreds of millions of years. During essentially all of that time the state is well described by H_0 alone: we can assign it energy levels, parities, magnetic moments, and so on. The *act* of decaying, once it happens, is comparatively instantaneous ($\sim 10^{-20}$ s). With up to ~ 20 orders of magnitude separating the lifetime from the decay-transition time, V' acts only very rarely and may be treated as a small perturbation.

Note:

There is an important exception, treated at the end of the lecture (Sec. 10.5): when an excited daughter level lies *accidentally close* in energy to the parent, the perturbative denominator blows up, V' is no longer negligible, and the true state really is a strong mixture of the two configurations.

10.2 Fermi's Golden Rule and the Density of States

Mathematical Setup

With $H = H_0 + V'$, time-dependent perturbation theory (Quantum Mechanics 2) gives the transition rate from a fixed initial state $|i\rangle$ to final states $|f\rangle$ as *Fermi's golden rule*:

Physical Law 10.1 Fermi's Golden Rule

The decay constant (transition rate) is

$$\lambda = \frac{2\pi}{\hbar} |V_{fi}|^2 \rho(E_f), \quad (156)$$

where $V_{fi} = \langle f|V'|i\rangle$ is the transition matrix element of the perturbation and $\rho(E_f)$ is the *density of final states* per unit energy.

Significance: the rate factorises into a *dynamical* part $|V_{fi}|^2$ (how strongly the interaction connects initial and final states) and a *kinematical / counting* part $\rho(E_f)$ (how many final states are available). Both must be large for a fast decay.

Physical Concept 10.1: Density of (Final) States

$$\rho(E_f) = \frac{dn_f}{dE_f}, \quad (157)$$

the number of accessible final states per unit energy at the energy E_f of the final configuration.

Physical Motivation — Why Count States?

Consider β -type decay $n \rightarrow p$ inside a nucleus. The emitted proton (or, more relevantly for β decay, the electron and neutrino) can come out with its momentum pointing in many different directions, each a distinct final state with the same energy. If several final states share the same energy, the rate is the sum over all of them — hence we multiply by the *number of states per unit energy*. Colloquially this available volume of final momenta is the *phase space* into which the system can decay.

Consequence 10.2 Phase Space Controls the Rate

The larger the phase space (the more final states available at the decay energy), the larger the decay probability, and the dependence is *linear*: $\lambda \propto \rho(E_f)$.

Significance: this is why β decay — with a three-body final state $(p, e, \bar{\nu})$ and therefore a large, continuously distributed phase space — requires careful phase-space bookkeeping, which we will do in detail when we reach it. Note that it is the *final*-state count that enters: the system starts in one definite initial state, so there is nothing to count on the initial side.

10.3 Finite Lifetime $\Rightarrow$ Lorentzian Energy Width

Physical Motivation

A truly stationary eigenstate of H_0 has a perfectly sharp energy. But a state that decays is *not* stationary — its probability leaks away in time. We now show that this time-dependence forces the state to have a *spread* of energies, with a characteristic Lorentzian line shape and width $\Gamma = \hbar/\tau$.

Step 1 — Build in the exponential decay. For a pure eigenstate of H_0 the wavefunction is $\Psi(\vec{r}, t) = \psi(\vec{r}) e^{-iEt/\hbar}$, whose modulus squared $|\Psi|^2 = |\psi(\vec{r})|^2$ is constant in time. That cannot describe decay. The empirical fact we must reproduce is the exponential decay law

$$|\Psi_a(\vec{r}, t)|^2 = |\Psi_a(\vec{r}, 0)|^2 e^{-t/\tau_a}, \quad (158)$$

where τ_a is the mean lifetime of state a . This is an *assumption*, imposed because it is what we observe. To produce (158), attach a real decaying envelope to the usual phase:

$$\boxed{\Psi_a(\vec{r}, t) = \psi_a(\vec{r}) e^{-iE_a t/\hbar} e^{-t/2\tau_a}, \quad t \geq 0.} \quad (159)$$

Significance: the factor $e^{-t/2\tau_a}$ (note the 2) is precisely what is needed so that squaring gives e^{-t/τ_a} ; it encodes the decay directly in the amplitude.

Step 2 — Transform to the energy domain. A state with definite energy oscillates as $e^{-iEt/\hbar}$. A state that is *not* a single energy eigenstate can be written as a continuous superposition — a Fourier transform — over energy:

$$\Psi_a(t) = \int_{-\infty}^{\infty} A(E) e^{-iEt/\hbar} dE. \quad (160)$$

(t and E are the conjugate Fourier pair, exactly like position and momentum.) Inverting and using the decay form (159), with decay starting at $t = 0$ (we may insert a step function $\theta(t)$; integrating from $-\infty$ would diverge):

$$A(E) \propto \int_0^{\infty} e^{+iEt/\hbar} e^{-iE_a t/\hbar} e^{-t/2\tau_a} dt = \int_0^{\infty} \exp\left[\left(\frac{i(E - E_a)}{\hbar} - \frac{1}{2\tau_a}\right)t\right] dt. \quad (161)$$

Step 3 — Do the elementary integral. The integrand is a single decaying exponential, so

$$A(E) \propto \left[\frac{\exp\left[\left(\frac{i(E-E_a)}{\hbar} - \frac{1}{2\tau_a}\right)t\right]}{\frac{i(E-E_a)}{\hbar} - \frac{1}{2\tau_a}} \right]_0^\infty = \frac{1}{\frac{1}{2\tau_a} - \frac{i(E-E_a)}{\hbar}}, \quad (162)$$

where the upper limit vanishes because the real part of the exponent is negative.

Step 4 — Square it. The probability of finding the system at energy E is $\propto |A(E)|^2$:

$$|A(E)|^2 \propto \frac{1}{\left(\frac{1}{2\tau_a}\right)^2 + \left(\frac{E-E_a}{\hbar}\right)^2} = \frac{\hbar^2}{(E-E_a)^2 + \left(\frac{\hbar}{2\tau_a}\right)^2}. \quad (163)$$

Defining the *energy width* $\Gamma_a \equiv \hbar/\tau_a$ and absorbing constants, the (unnormalised) probability density in energy is

Physical Concept 10.2: Lorentzian Line Shape of a Decaying Level

$$P(E) dE = \frac{dE}{(E-E_a)^2 + \frac{\Gamma_a^2}{4}}, \quad \Gamma_a \equiv \frac{\hbar}{\tau_a}. \quad (164)$$

This is a *Lorentzian* centred at the level energy E_a with full width at half maximum (FWHM) equal to Γ_a .

Significance: a decaying state does *not* have a sharp energy — its energy is smeared into a Lorentzian whose width is inversely proportional to the lifetime. An infinitely long-lived state ($\tau \rightarrow \infty$) has $\Gamma \rightarrow 0$, recovering a perfectly sharp δ -function level.

Check: Γ is the FWHM. The maximum of (164) is at $E = E_a$, value $4/\Gamma_a^2$. Setting $E - E_a = \Gamma_a/2$ gives $1/\left(\frac{\Gamma_a^2}{4} + \frac{\Gamma_a^2}{4}\right) = 2/\Gamma_a^2$, exactly half the maximum — so the full width between the half-maximum points is Γ_a .

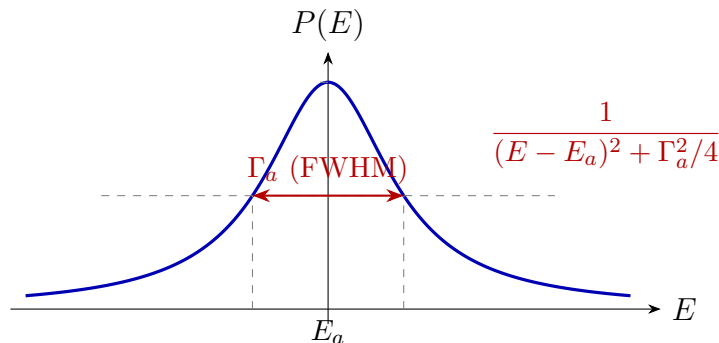


Figure 7: Lorentzian energy distribution of a decaying level. The peak sits at the level energy E_a ; the full width at half maximum equals the energy width $\Gamma_a = \hbar/\tau_a$. A longer-lived state is narrower; a stable state ($\tau \rightarrow \infty$) is a δ -spike.

Physical Intuition — The Energy–Time Uncertainty Relation

This result is the energy–time uncertainty relation in disguise. A pure tone (single frequency) is perfectly defined in frequency but has no localisation in time; to build a pulse that is localised in time you must superpose a *range* of frequencies. Translating frequency $\rightarrow$ energy ($E = \hbar\omega$) and pulse duration $\rightarrow$ lifetime, broadening the energy distribution shortens the lifetime:

$$\boxed{\Delta E \cdot \Delta t \sim \hbar \iff \Gamma \tau = \hbar.} \quad (165)$$

Significance: any level that can decay in a finite time is necessarily energy-broadened; a quoted level energy is only as sharp as the level’s lifetime allows.

Worked Example 10.1 (How Big Is the Width? (Why We Ignore It in the DOS))

Using $\hbar = 6.58 \times 10^{-16} \text{ eV}\cdot\text{s}$:

- Lifetime $\tau = 1 \text{ s}$: $\Gamma = \hbar/\tau \approx 6.6 \times 10^{-16} \text{ eV}$.
- Lifetime $\tau = 1 \text{ fs} = 10^{-15} \text{ s}$ (typical of a fast nuclear excited state): $\Gamma \approx 0.66 \text{ eV}$.

Even for very short-lived nuclear states the energy width is $\lesssim \text{eV}$ — utterly negligible compared with the MeV-scale spacing of nuclear levels.

Significance: because the *energy* width of the final state is so tiny relative to nuclear level spacings, we do *not* need to fold it into $\rho(E_f)$. The density of states that matters in Eq. (156) is the *momentum / position* phase space of the final particles, not the intrinsic energy width of the final level.

10.4 Partial Widths and Branching Ratios

Physical Motivation

A single state often has more than one way to decay — e.g. state A may go to B 80% of the time and to C 20% of the time. Each channel proceeds at its own rate, and the total decay rate is their sum. Since rate, width and inverse-lifetime are all proportional ($\Gamma = \hbar/\tau$, $\lambda = 1/\tau$), the widths add.

Physical Law 10.3 Additivity of Partial Widths

If a state decays through several channels i , the total width is the sum of the partial widths,

$$\boxed{\Gamma = \sum_i \Gamma_i,} \quad (166)$$

and the fraction going through channel i — the *branching ratio* — is

$$\boxed{\text{BR}_i = \frac{\Gamma_i}{\Gamma} = \frac{\lambda_i}{\lambda}.} \quad (167)$$

Significance: the total width (hence total lifetime) and the branching ratios are independently measurable, so the *partial* widths $\Gamma_i = \text{BR}_i \Gamma$ can be extracted. This is how one separately measures, say, the partial width for γ decay versus α decay of the same

level — two experiments (lifetime + branching fractions) combine to reveal the level's decay structure.

10.5 Accidental Near-Degeneracy Enhancement

Mathematical Setup

First-order perturbation theory corrects the parent state ψ_a by admixing the daughter state ψ_b :

$$\psi = \psi_a + \frac{V'_{ba}}{E_a - E_b} \psi_b, \quad (168)$$

where $V'_{ba} = \langle b|V'|a\rangle$ and E_a, E_b are the unperturbed energies of parent and daughter.

When the perturbation stops being small. Normally V'_{ba} is tiny, so the admixture is negligible. But if — by accident — an *excited* daughter level lies almost exactly at the parent energy, $E_a \approx E_b$, the denominator $E_a - E_b \rightarrow 0$ and the coefficient is strongly enhanced. The supposedly small V' then no longer dominates over nothing; the true state becomes a near-equal mixture of ψ_a and ψ_b that can oscillate between the two configurations.

Note:

In this regime the clean “ H_0 plus tiny V' ” picture (Sec. 10.1) breaks down, and the decay probability is correspondingly much larger. Such accidental degeneracies are responsible for anomalously fast transitions.

10.6 Alpha Decay: Definition, Discovery, and Q -value

Physical Motivation

Historically, α radiation was identified as the least-penetrating of the three classical radiations (shortest range — as we saw in Lecture 6 when discussing radiation–matter interaction), and its nature as ${}^4\text{He}$ nuclei was confirmed by letting a radioactive source decay into a sealed vessel and later detecting the accumulated helium gas spectroscopically.

Physical Concept 10.3: Alpha Decay

A heavy nucleus emits a ${}^4\text{He}$ nucleus (the α particle):



The daughter Y has 2 fewer protons and 2 fewer neutrons.

Significance: α decay moves a nucleus by $(\Delta Z, \Delta N) = (-2, -2)$ on the chart of nuclides — the “yellow” step encountered in Lecture 5 — and is the dominant decay mode for the heaviest nuclei ($A \gtrsim 200$).

Physical Concept 10.4: Q -value of Alpha Decay

$$Q_\alpha = [M({}_Z^AX) - M({}_{Z-2}^{A-4}Y) - M({}_2^4\text{He})]c^2. \quad (170)$$

Spontaneous decay requires $Q_\alpha > 0$.

Significance: Q_α is the energy released, sourced entirely by the change in total binding energy; it is the energy budget shared (almost entirely) by the α particle as kinetic energy.

Why α and Not Other Light Fragments?

A natural question: why is ${}^4\text{He}$ emitted rather than a proton, neutron, deuteron, triton, ${}^3\text{He}$, or ${}^{12}\text{C}$? The naive answer “because helium is a noble gas” is *wrong* — noble-gas chemistry concerns atomic electrons ($\sim \text{eV}$ binding), which are completely irrelevant to a nuclear process at the MeV scale. The real reason is the *exceptional binding* of ${}^4\text{He}$:

- ${}^4\text{He}$ is doubly magic ($Z = N = 2$, both closed shells), making it unusually tightly bound — a local peak on the B/A curve at $A = 4$.
- Computing the Q -value for emitting p , n , d , t , ${}^3\text{He}$, ${}^6\text{Li}$, ... from a heavy nucleus (e.g. ${}^{232}\text{U}$) gives $Q < 0$ for all of them: those emissions are *endothermic* and cannot happen spontaneously. Only α emission (and, far more rarely, heavier clusters such as ${}^{12}\text{C}$) gives $Q > 0$.
- Bonus from pairing: removing an even–even α leaves the even–even structure of the parent favourable.

Note:

Heavier-cluster decays (e.g. ${}^{12}\text{C}$ emission) can be energetically allowed but are observed with branching ratios many orders of magnitude smaller than α ; the tunnelling theory below explains why heavier, more highly-charged fragments are so strongly suppressed.

10.7 Alpha Decay Kinematics

Mathematical Setup — Conservation Laws

Is Q_α entirely the α ’s kinetic energy? Almost — but the daughter must recoil to conserve momentum, so it carries a small share. Work in the rest frame of the parent (total momentum zero).

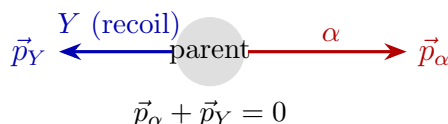


Figure 8: Two-body decay in the parent rest frame: the α and the recoiling daughter fly back-to-back with equal and opposite momenta, $|\vec{p}_\alpha| = |\vec{p}_Y| \equiv p$.

Momentum conservation gives $|\vec{p}_\alpha| = |\vec{p}_Y| \equiv p$ (back to back). Energy conservation (non-relativistic, since $T \ll mc^2$ here):

$$Q_\alpha = T_\alpha + T_Y = \frac{p^2}{2m_\alpha} + \frac{p^2}{2m_Y} = T_\alpha \left(1 + \frac{m_\alpha}{m_Y} \right). \quad (171)$$

Solving for each kinetic energy:

$$T_\alpha = Q_\alpha \frac{m_Y}{m_\alpha + m_Y}, \quad T_Y = Q_\alpha \frac{m_\alpha}{m_\alpha + m_Y}. \quad (172)$$

Significance: the α carries the lion's share; the heavy daughter takes only a small recoil energy. Using $m_\alpha \approx 4u$ and $m_Y \approx (A-4)u$,

$$T_\alpha \approx Q_\alpha \left(1 - \frac{4}{A}\right), \quad T_Y \approx Q_\alpha \frac{4}{A}. \quad (173)$$

Significance: for $A \gtrsim 200$ the daughter recoil is only a couple of percent of Q_α , so the measured α energy is very nearly Q_α — giving discrete, characteristic α energies for each emitter.

Note:

Although the nucleon *count* balances exactly ($A = (A-4) + 4$), the Q -value is non-zero because it lives in the *binding-energy* differences: the rearranged configuration has lower total mass (higher total binding) when $Q_\alpha > 0$.

10.8 The Geiger–Nuttall Law (Empirical)

Physical Motivation

Two striking empirical regularities of α decay, noticed early, must be reproduced by any successful theory. The first is the *Geiger–Nuttall law*: within an isotopic (single- Z) chain, the half-life depends extraordinarily strongly on the α energy.

Physical Law 10.4 Geiger–Nuttall Law

For a fixed daughter charge, the logarithm of the half-life is linear in $1/\sqrt{Q_\alpha}$:

$$\log_{10} t_{1/2} = \frac{a(Z)}{\sqrt{Q_\alpha}} + b(Z), \quad (174)$$

with element-dependent constants $a(Z)$, $b(Z)$.

Significance: a modest change in Q_α produces an *exponential* change in lifetime — half-lives across the α -emitters span from microseconds to billions of years. As Q_α increases, the half-life plummets; as Q_α decreases, it soars.

Second regularity: the A -dependence. Along an isotopic chain, plotting Q_α (equivalently E_α) versus mass number A shows a smooth trend. Combining the two regularities: moving to the right on the chart of nuclides within α -emitters, Q_α decreases and hence the half-life increases (until β decay takes over and ends the trend). Both must emerge from one underlying theory.

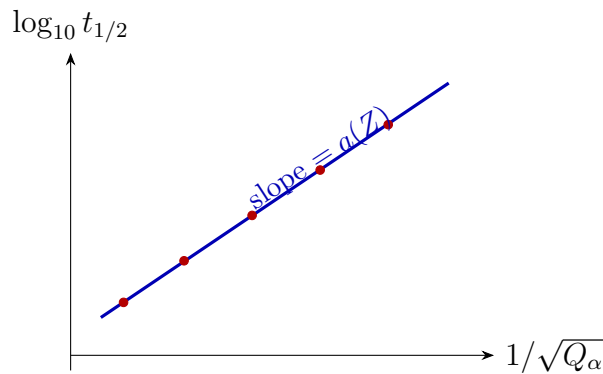


Figure 9: Geiger–Nuttall law for a single isotopic chain: $\log_{10} t_{1/2}$ is linear in $1/\sqrt{Q_\alpha}$. Smaller Q_α (right) $\Rightarrow$ longer half-life. The huge dynamic range ($\sim \mu\text{s}$ to Gyr) reflects the exponential sensitivity derived below.

10.9 Gamow Theory: Tunnelling Through the Coulomb Barrier

Physical Motivation

The model (Gamow): inside a large nucleus, two protons and two neutrons may transiently cluster into a pre-formed α particle. This α is bound inside the short-range nuclear potential well, but outside the nuclear radius it feels only the long-range Coulomb repulsion of the daughter. Classically it is trapped; quantum-mechanically it can *tunnel* through the Coulomb barrier and escape.

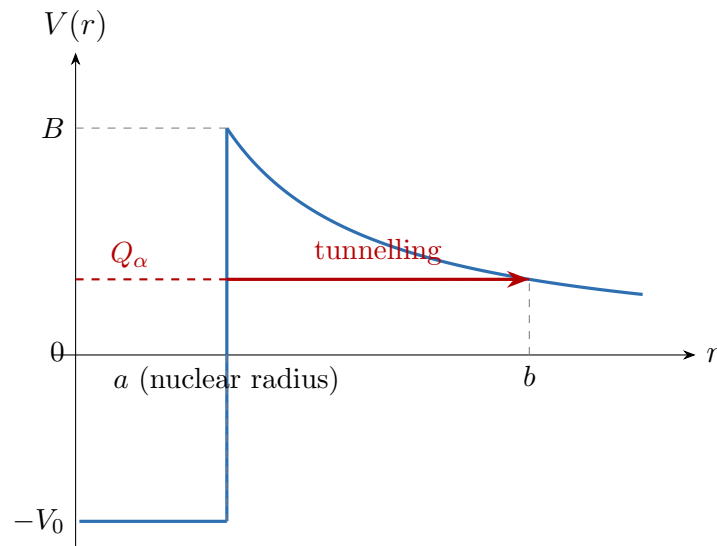


Figure 10: Potential energy seen by the α particle. Inside the nuclear radius a it sits in a deep well of depth V_0 ; outside, the Coulomb repulsion falls as $1/r$. The barrier reaches height B at $r = a$ and drops to the α energy Q_α at the outer turning point $r = b$. Classically the α (energy Q_α) is trapped between a and b ; quantum mechanically it tunnels through this region and escapes.

Mathematical Setup — Barrier Geometry

Define two key radii and the barrier height. The Coulomb potential between the α (charge $2e$) and daughter (charge $Z_d e$, where $Z_d = Z - 2$) is

$$V(r) = \frac{1}{4\pi\epsilon_0} \frac{2Z_d e^2}{r}. \quad (175)$$

- **Barrier height** at the nuclear surface $r = a$:

$$B = \frac{1}{4\pi\epsilon_0} \frac{2Z_d e^2}{a}. \quad (176)$$

- **Outer turning point** b , where the Coulomb potential equals the α energy, $V(b) = Q_\alpha$:

$$b = \frac{1}{4\pi\epsilon_0} \frac{2Z_d e^2}{Q_\alpha}. \quad (177)$$

The height the α must overcome to reach the outside world is exactly $B - Q_\alpha$, and dividing (176) by (177) gives the convenient ratio

$$\frac{a}{b} = \frac{Q_\alpha}{B}. \quad (178)$$

Mathematical Setup — WKB Tunnelling Through a Varying Barrier

For a *square* barrier the transmission probability decays exponentially with the imaginary wavenumber $k_2 = \sqrt{2m(V - E)}/\hbar$. For our *smoothly varying* Coulomb barrier, slice it into many thin square barriers of width dr . The infinitesimal transmission through one slice is

$$dP = \exp \left[-2 dr \sqrt{\frac{2m_\alpha}{\hbar^2} (V(r) - Q_\alpha)} \right], \quad (179)$$

and the total transmission is the product over slices, i.e. the exponential of the integral:

$$P = e^{-2G}, \quad G = \sqrt{\frac{2m_\alpha}{\hbar^2}} \int_a^b \sqrt{V(r) - Q_\alpha} dr. \quad (180)$$

Significance: G — the *Gamow factor* — is the accumulated exponential suppression of tunnelling; it controls the entire enormous range of α half-lives.

Evaluating the Gamow Integral

Carrying out the integral (180) with the Coulomb form (175) gives the closed-form result

$$G = \frac{2Z_d e^2}{4\pi\epsilon_0 \hbar} \sqrt{\frac{2m_\alpha}{Q_\alpha}} \left[\cos^{-1} \sqrt{\frac{a}{b}} - \sqrt{\frac{a}{b}} \sqrt{1 - \frac{a}{b}} \right]. \quad (181)$$

Significance: this single expression contains both empirical laws. The bracket depends only on the ratio $a/b = Q_\alpha/B$, while the prefactor carries the explicit $1/\sqrt{Q_\alpha}$ dependence.

Thick-barrier (Geiger–Nuttall) limit. When $Q_\alpha \ll B$ (a tall, thick barrier, as for real heavy emitters), $a/b \rightarrow 0$, so $\cos^{-1} \sqrt{a/b} \rightarrow \pi/2$ and the second term vanishes. Then

$$G \approx \frac{\pi Z_\alpha Z_d e^2}{4\pi\epsilon_0 \hbar v_\alpha}, \quad v_\alpha = \sqrt{\frac{2Q_\alpha}{m_\alpha}}, \quad Z_\alpha = 2, \quad (182)$$

so that $G \propto Z_d/\sqrt{Q_\alpha}$. Since the decay rate carries the factor e^{-2G} , we have $\ln \lambda \propto -2G \propto -Z_d/\sqrt{Q_\alpha}$ — *precisely* the Geiger–Nuttall form (174). *Significance:* the Coulomb-barrier tunnelling integral reproduces the exponential $1/\sqrt{Q_\alpha}$ law that was discovered empirically decades earlier.

From Tunnelling Probability to Decay Rate

The transmission P is the chance of escaping in a *single* attempt. The α rattles back and forth inside the well, hitting the barrier with an *assault frequency*

$$f \approx \frac{v_\alpha}{2a}, \quad (183)$$

where v_α is the internal α velocity and $2a$ the nuclear diameter. Numerically $f \sim 10^{22} \text{ s}^{-1}$. The decay constant is the product

$$\lambda = f P = \frac{v_\alpha}{2a} e^{-2G}. \quad (184)$$

Significance: this simple “frequency $\times$ tunnelling probability” estimate predicts α half-lives that, while sensitive to the poorly-known nuclear radius a , come out remarkably close to experiment over ~ 20 orders of magnitude — a triumph of the quantum tunnelling picture.

Note:

Two simplifications were made: (i) the orbital angular momentum was taken to be $\ell = 0$, and (ii) the nucleus was assumed spherical. Both corrections matter quantitatively; deformed nuclei have direction-dependent radii (easier escape along the long axis), and $\ell \neq 0$ adds a centrifugal barrier, treated next.

10.10 Angular-Momentum (Centrifugal) Barrier

Conceptual Background

So far the barrier was purely the Coulomb potential. But if the α is emitted with orbital angular momentum $\ell \neq 0$, the radial Schrödinger equation acquires an extra repulsive term. Recall that separating the 3D Schrödinger equation into radial and angular parts produces, in the radial equation, the *centrifugal* term

$$V_\ell(r) = \frac{\hbar^2 \ell(\ell+1)}{2\mu r^2}, \quad (185)$$

where μ is the reduced mass. Adding it to the Coulomb potential defines an *effective potential* $V_{\text{eff}}(r) = V(r) + V_\ell(r)$. *Significance:* angular momentum raises the barrier the α must tunnel through, so higher ℓ *suppresses* the decay.

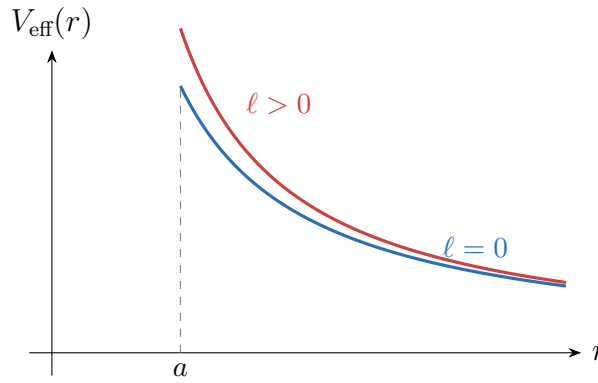


Figure 11: Effective potential for $\ell = 0$ (Coulomb only, blue) and $\ell > 0$ (Coulomb + centrifugal, red). The centrifugal term $\hbar^2\ell(\ell + 1)/2\mu r^2$ pushes the barrier higher and outward, increasing the tunnelling integral G and thus lowering the decay probability.

Physical intuition (off-centre emission). Emitting the α from the *side* of the nucleus rather than radially requires applying a torque $\vec{\tau} = \vec{r} \times \vec{F}$ to give it angular momentum — it is “harder” to throw the α sideways than straight out. Equivalently, a larger ℓ means a higher effective barrier, a larger G , and an exponentially smaller escape probability.

10.11 Selection Rules for Alpha Decay

Conceptual Background

A transition between two nuclear states must conserve total angular momentum and parity. For α decay these conservation laws translate into selection rules on the orbital angular momentum ℓ carried by the emitted α .

Only orbital angular momentum is carried. The α particle (${}^4\text{He}$) has spin–parity $J^\pi = 0^+$: its two protons and two neutrons couple to zero spin, and its intrinsic parity is $+$. Hence the α cannot carry away any *intrinsic* (spin) angular momentum — only *orbital* angular momentum ℓ . Therefore the entire change in nuclear spin must be supplied by ℓ .

Physical Law 10.5 Alpha-Decay Selection Rules

For a transition from parent spin–parity $I_i^{\pi_i}$ to daughter $I_f^{\pi_f}$, the emitted α carries orbital angular momentum ℓ with

$$|I_i - I_f| \leq \ell \leq I_i + I_f \quad (186)$$

(vector addition of angular momenta), and the parities must satisfy

$$\pi_i = \pi_f (-1)^\ell. \quad (187)$$

Significance: the parity rule ties the allowed ℓ values to the relative parity of parent and daughter; if $\pi_i = \pi_f$ then ℓ must be even, and if $\pi_i = -\pi_f$ then ℓ must be odd. Combined with the centrifugal barrier (Sec. 10.10), *larger required ℓ means a more strongly hindered (slower) decay.*

Worked Example 10.2 ($^{232}\text{Th} \rightarrow ^{228}\text{Ra}$ Ground-State Transition)

$^{232}_{90}\text{Th} \rightarrow ^{228}_{88}\text{Ra} + \alpha$. Both nuclei are even–even, so their ground states are $I^\pi = 0^+$.

Angular momentum: $|0 - 0| \leq \ell \leq 0 + 0 \Rightarrow \ell = 0$.

Parity: $\pi_i = \pi_f(-1)^\ell \Rightarrow (+) = (+)(-1)^0 = (+)$. ✓

The ground-state-to-ground-state transition is *allowed* with $\ell = 0$ — no centrifugal hindrance. (Transitions to excited $0^+, 2^+, \dots$ daughter levels and their decay modes are taken up at the start of the next lecture.)

Key Takeaways

- **Decay is dynamical, not static:** split $H = H_0 + V'$ with $V' \ll H_0$ (separation of time scales) rather than writing a static superposition.
- **Fermi's golden rule:** $\lambda = \frac{2\pi}{\hbar} |V_{fi}|^2 \rho(E_f)$; the rate scales *linearly* with the available final-state phase space $\rho(E_f) = dn_f/dE_f$.
- **Energy width:** a finite lifetime τ gives a Lorentzian level of FWHM $\Gamma = \hbar/\tau$ — the energy–time uncertainty relation. The width is tiny ($\lesssim \text{eV}$), so only *momentum* phase space matters in the DOS.
- **Partial widths add:** $\Gamma = \sum_i \Gamma_i$, $\text{BR}_i = \Gamma_i/\Gamma$ — lets one extract channel-by-channel rates.
- **Alpha decay:** $^A_Z\text{X} \rightarrow ^{A-4}_{Z-2}\text{Y} + \alpha$; emitted because ^4He is doubly magic (large $Q_\alpha > 0$), not because it is a noble gas. $T_\alpha \approx Q_\alpha(1 - 4/A)$.
- **Geiger–Nuttall** $\log_{10} t_{1/2} \propto 1/\sqrt{Q_\alpha}$ follows from **Gamow tunnelling:** $P = e^{-2G}$, $G \propto Z_d/\sqrt{Q_\alpha}$ in the thick-barrier limit; $\lambda = (v_\alpha/2a)e^{-2G}$.
- **Angular momentum** adds a centrifugal barrier $\hbar^2\ell(\ell+1)/2\mu r^2$; higher ℓ hinders decay. Selection rules: $|I_i - I_f| \leq \ell \leq I_i + I_f$ and $\pi_i = \pi_f(-1)^\ell$.

11 Lecture 09 — Beta Decay: Fermi Theory, Spectrum & Neutrinos

Lecture date: 9 June 2026 — *Lecturer:* Dr. Moshe Friedman.

Overview

Having treated α decay in Lecture 8, we now turn to the second great decay channel: *beta decay*, the manifestation of the *weak interaction* in nuclei. The lecture builds the *Fermi theory* of β decay (1933) from the ground up: starting again from Fermi’s golden rule, we factorise the transition rate into a weak coupling, a nuclear matrix element, and a *lepton phase-space* factor. Counting the available states of the emitted electron and antineutrino reproduces the characteristic *continuous* β spectrum $N(p) \propto p^2(Q - T_e)^2$ — historically the very anomaly that forced Pauli to postulate the neutrino. We then add the Coulomb correction (the Fermi function), define the experimentally central quantity $ft_{1/2}$ and its logarithm $\log ft$, classify transitions (Fermi, Gamow–Teller, forbidden, super-allowed) via selection rules, and close with three landmark pieces of neutrino physics: the endpoint measurement of the neutrino mass, neutrino helicity, and the solar neutrino problem as evidence for neutrino oscillations (hence mass). A brief statement of the γ -decay multipole selection rules and a real ^{23}Al decay-spectroscopy study tie the chapter together.

11.1 The Three Modes of Beta Decay and the Valley of Stability

Physical Motivation

On the chart of nuclides, stable isotopes trace a narrow curved band — the *valley (or line) of stability*. Think of it *topographically*: the binding-energy landscape is a wadi (a dry riverbed). A nucleus off the floor of the valley “rolls downhill” toward stability, but it can only move *along* isobaric ($A = \text{const}$) chains, never across them, because the decays that drive it conserve mass number. The agent that walks a nucleus diagonally back into the valley is *beta decay*.

Physical Concept 11.1: The Three Beta Decay Modes

All three conserve mass number A and shift the nucleus by one unit in Z :

$$\beta^- : \quad n \rightarrow p + e^- + \bar{\nu}_e, \quad (Z, N) \rightarrow (Z + 1, N - 1), \quad (188)$$

$$\beta^+ : \quad p \rightarrow n + e^+ + \nu_e, \quad (Z, N) \rightarrow (Z - 1, N + 1), \quad (189)$$

$$\text{EC} : \quad p + e^- \rightarrow n + \nu_e, \quad (Z, N) \rightarrow (Z - 1, N + 1). \quad (190)$$

Here β^- acts on the neutron-rich (right) flank of the valley, β^+ on the proton-rich (left) flank. *Electron capture* (EC) absorbs an *atomic* inner-shell electron — one that spends part of its time inside the nucleus — and yields the same nuclear daughter as β^+ .

Conceptual Background — Conservation Laws

A bare $n \rightarrow p$ conversion would violate charge conservation, so a charged lepton must accompany it; and a bare $n \rightarrow p + e^-$ would violate *lepton number* conservation, so a neutral lepton (a neutrino) is also required. Assigning lepton number $L = +1$ to leptons

(e^-, ν) and $L = -1$ to antileptons $(e^+, \bar{\nu})$:

$$\boxed{\Delta Q = 0 \quad (\text{charge}), \quad \Delta L = 0 \quad (\text{lepton number}).} \quad (191)$$

Significance: these two bookkeeping rules *alone* dictate which neutral lepton appears — a $\bar{\nu}_e$ accompanies the e^- in β^- , while a ν_e accompanies the e^+ in β^+ and the absorbed e^- in EC.

Note:

There are *three generations* of leptons, e, μ, τ , each with its own neutrino ν_e, ν_μ, ν_τ . Nuclear β decay involves only the electronic flavour, so throughout we write simply $\nu, \bar{\nu}$ for $\nu_e, \bar{\nu}_e$.

Why β Dominates Over α Off the Heavy Corner

α decay moves a nucleus two steps down and two left ($\Delta Z = -2, \Delta N = -2$) — also diagonally toward the valley — but is favoured only at very high A . Elsewhere β decay dominates because (i) for β^- there is *no Coulomb barrier* (the emitted electron is attracted, not repelled, by the daughter), and (ii) the leptons do not feel the strong force that confines nucleons, so the process is intrinsically simpler and proceeds whenever it is energetically allowed. Astrophysically, violent neutron-rich environments (supernovae, neutron-star mergers, the r -process) and reactor fission both create nuclei far from stability that then β -decay back — making β -decay theory central to nucleosynthesis and reactor design.

β^+ versus EC — an energetics argument. β^+ and EC give the *same* nuclear daughter, so which wins? The decisive point is the rest-mass budget. In β^+ a positron must be *created from nothing* ($+m_e c^2$) and the captured/lost electron must be accounted for ($+m_e c^2$ relative to EC, which instead *consumes* an already-present electron). Hence β^+ costs about

$$\boxed{Q_{\beta^+} = Q_{\text{EC}} - 2m_e c^2, \quad 2m_e c^2 = 1.022 \text{ MeV}.} \quad (192)$$

Significance: EC is *always* open if β^+ is, but not conversely — a nucleus may have just enough energy for EC yet too little (< 1.022 MeV margin) for β^+ . This is why chart entries often list a combined symbol ε (EC) or ε/β^+ as competing channels on the proton-rich side. EC is harder to detect (only an X-ray from the refilling atomic shell), whereas β^+ ejects a measurable positron.

11.2 Fermi Theory: Golden Rule and the Matrix Element

Conceptual Background

Fermi theory (mid-1930s) predates and is not fully consistent with the modern Standard Model, but it is more than adequate for nuclear physics. As in Lecture 8 we start from *Fermi's golden rule* for the decay constant,

$$\lambda = \frac{2\pi}{\hbar} |V_{fi}|^2 \rho(E_f), \quad (193)$$

and our task is to evaluate the two factors — the transition matrix element V_{fi} and the density of final states $\rho(E_f)$ — for the specific case where the final state contains the daughter nucleus *plus* a created electron and antineutrino.

Mathematical Setup — the Transition Matrix Element

By definition $V_{fi} = \int \psi_f^* V \psi_i dV$. Fermi's ansatz writes the weak interaction as a point “contact” coupling of strength G , with the final state a product of three wavefunctions — daughter nucleus, electron, antineutrino:

$$V_{fi} = G \int \psi_f^* \phi_e^* \phi_{\bar{\nu}}^* \mathcal{O}(x) \psi_i dV, \quad (194)$$

where ψ_i, ψ_f are the initial/final *nuclear* states, $\phi_e, \phi_{\bar{\nu}}$ the lepton wavefunctions, and $\mathcal{O}(x)$ the (so far unknown) interaction operator. The leptons are created as free particles, so we model them as plane waves normalised in a box of volume V :

$$\phi_e = \frac{1}{\sqrt{V}} e^{-i\mathbf{p}\cdot\mathbf{r}/\hbar}, \quad \phi_{\bar{\nu}} = \frac{1}{\sqrt{V}} e^{-i\mathbf{q}\cdot\mathbf{r}/\hbar}, \quad (195)$$

with $\mathbf{p}$ the electron momentum and $\mathbf{q}$ the antineutrino momentum.

The Allowed Approximation

Expand each lepton plane wave in a Taylor series about the nucleus:

$$e^{-i\mathbf{p}\cdot\mathbf{r}/\hbar} = 1 - \frac{i\mathbf{p}\cdot\mathbf{r}}{\hbar} + \dots \quad (196)$$

The size of the first correction is fixed by physics we already know: the electron carries $p \sim 1 \text{ MeV}/c$ and is emitted from *within* the nucleus, $r \lesssim R \sim 1 \text{ fm}$ (“the electron leaves the nucleus, not the Moon”). Then

$$\frac{pR}{\hbar} \sim \frac{(1 \text{ MeV}/c)(1 \text{ fm})}{\hbar} \approx 5 \times 10^{-3} \ll 1, \quad (197)$$

so we keep only the leading term, $e^{-i\mathbf{p}\cdot\mathbf{r}/\hbar} \approx 1$, and likewise for the antineutrino.

Physical Concept 11.2: Allowed Approximation

Two assumptions taken together:

- (i) replace both lepton plane waves by unity inside the nucleus ($pR/\hbar \ll 1$, equivalently $\Delta L = 0$);
- (ii) the surviving *nuclear* matrix element $M_{fi} = \int \psi_f^* \psi_i dV$ is taken *independent* of the lepton momenta $\mathbf{p}, \mathbf{q}$ (a constant pulled out of the phase-space integral).

Significance: under (196)–(197) the two normalisation factors $1/\sqrt{V}$ survive as $1/V$, the operator sandwich collapses to a nuclear overlap M_{fi} , and *all* the momentum dependence of λ is pushed into the density of states. The leptons are created uniformly likely everywhere inside the nucleus.

11.3 Lepton Phase Space and the Density of States

Mathematical Setup

The final state is (discrete nuclear level) $\times$ (electron) $\times$ (antineutrino). Because nuclear levels are essentially discrete on the scale of the lepton continuum, the nuclear density

of states is ≈ 1 ; all the counting is in the *lepton* momenta. Count states in phase space (position $\times$ momentum), with the smallest phase-space cell h^3 :

$$\text{spatial factor} = \frac{V}{h^3} \quad (\text{one for } e, \text{ one for } \bar{\nu}). \quad (198)$$

For the momentum part, the number of states with $|\mathbf{p}|$ in $[p, p + dp]$ is the volume of a spherical shell in momentum space (since $p^2 = p_x^2 + p_y^2 + p_z^2$):

$$dn_e = \frac{4\pi p^2 dp V}{h^3}, \quad dn_{\bar{\nu}} = \frac{4\pi q^2 dq V}{h^3}. \quad (199)$$

The joint number of final lepton states is therefore

$$dn_e dn_{\bar{\nu}} = \frac{(4\pi)^2 V^2 p^2 q^2 dp dq}{h^6}. \quad (200)$$

Folding It All Together

Insert $|V_{fi}|^2$ (which gives $G^2|M_{fi}|^2/V^2$ — the two box volumes *cancel* the V^2 above, so no choice of normalisation volume survives) and the density of states (200) into the golden rule (193). The differential decay constant for fixed lepton momenta is

$$d\lambda = \frac{2\pi}{\hbar} G^2 |M_{fi}|^2 \frac{(4\pi)^2 p^2 q^2 dp dq}{h^6}. \quad (201)$$

We want the rate *per unit energy* of the final nuclear level, so we must trade dq for dE_f . Assuming the neutrino is *massless*, its total energy is $E_{\bar{\nu}} = qc$; with total released energy split between the two leptons, $E_f = E_e + qc$ at fixed E_e , so

$$\boxed{\frac{dq}{dE_f} = \frac{1}{c}}. \quad (202)$$

Eliminating q via Energy Conservation

The two momenta are linked by energy conservation. The total kinetic energy released, Q (the Q -value), is shared between electron and antineutrino, so the antineutrino momentum is

$$\boxed{q = \frac{Q - T_e}{c}}, \quad (203)$$

where T_e is the electron kinetic energy. Substituting (202)–(203) into (201) and collecting constants yields the *momentum distribution* of the emitted electrons.

Physical Law 11.1 Beta Spectrum (Allowed, Uncorrected)

$$\boxed{N(p) dp \propto p^2 (Q - T_e)^2 dp = p^2 q^2 c^2 dp.} \quad (204)$$

Significance: the continuous β spectrum is *pure phase space*. The two factors are the electron's own state count (p^2) and the antineutrino's ($q^2 \propto (Q - T_e)^2$); their product peaks in the middle and vanishes at both endpoints, exactly the bell-skewed shape seen experimentally.

Physical Intuition — Why a Continuous Spectrum?

A two-body decay gives the daughter a *fixed* momentum; a single discrete γ would have a sharp line. Here, three bodies share the energy, so the electron's momentum is *not* fixed. The shape is a pure counting effect, in exact analogy with rolling dice:

- One die: each face 1–6 equally likely (uniform).
- Two dice: the sum is peaked at 7 *not* because nature “prefers” 7, but because 7 has *six* realisations (1+6, ..., 6+1) while 2 has only one.

The momentum is a *vector*; we measure only its magnitude, so the number of ways to realise a given $|\mathbf{p}|$ grows as $4\pi p^2$. With two leptons sharing a fixed total energy, the joint count $p^2 q^2$ is maximal when the energy is shared (the centre of the distribution), and zero when one lepton takes everything. This is the same many-states-dominate logic as the entropy maximum in statistical mechanics, here a counting on a six-dimensional ($\mathbf{p} \times \mathbf{q}$) phase space.

Note:

Pitfall: the spectrum is *not* a thermal or dynamical preference. Each microstate is equally probable; the peak arises purely because more microstates realise an intermediate momentum. Do not invoke any “force” that pushes the electron to mid-spectrum.

11.4 Coulomb Correction: the Fermi Function

Physical Motivation

Comparing the predicted $N(p)$ to data for ^{64}Cu — which decays both by β^- (to ^{64}Zn , 38%) and by β^+/EC (to ^{64}Ni , 61%) — the β^+ branch is visibly *shifted to higher momenta*. The cause: our counting assumed all final states equally probable, ignoring the electromagnetic interaction between the *charged* lepton and the daughter nucleus. A positron is *repelled* by the positively charged daughter (a forward “kick”, depleting low momenta); an electron is *attracted* (enhancing low momenta).

Physical Concept 11.3: Fermi Function $F(Z', p)$

A multiplicative correction depending only on the *daughter* charge Z' and the lepton momentum p (or energy E), accounting for the Coulomb distortion of the outgoing lepton wavefunction:

$$N(p) dp \propto F(Z', p) p^2 (Q - T_e)^2 |M_{fi}|^2 S(p, q) dp. \quad (205)$$

Significance: $F(Z', p)$ restores agreement with experiment; $F > 1$ enhances low-energy electrons (β^-), $F < 1$ suppresses low-energy positrons (β^+). The extra factor $S(p, q)$ is the *shape factor*: in the allowed approximation $S = 1$, but in general the matrix element may depend on p, q and $S \neq 1$ (forbidden transitions).

11.5 Total Rate, the ft Value and $\log ft$

Mathematical Setup

The total decay constant integrates the corrected spectrum over all allowed electron momenta, from 0 to the maximum $p_{\max}$ (set by $T_e = Q$):

$$\lambda = \frac{G^2 |M_{fi}|^2}{2\pi^3 \hbar^7 c^3} \int_0^{p_{\max}} F(Z', p) p^2 (Q - T_e)^2 dp. \quad (206)$$

Define the dimensionless *Fermi integral* $f(Z', E_0)$ by extracting the electron mass scales ($E_0 \equiv$ endpoint total energy, $E \equiv$ electron total energy):

$$f(Z', E_0) = \frac{1}{(m_e c)^3 (m_e c^2)^2} \int_0^{p_{\max}} F(Z', p) p^2 (E_0 - E)^2 dp. \quad (207)$$

The function f is *calculable*; the half-life $t_{1/2}$ is *measurable* via $\lambda = \ln 2 / t_{1/2}$ with $\ln 2 \approx 0.693$.

Physical Law 11.2 The Comparative Half-Life $ft_{1/2}$

$$ft_{1/2} = \frac{0.693 (2\pi^3 \hbar^7)}{G^2 m_e^5 c^4 |M_{fi}|^2}. \quad (208)$$

Significance: $ft_{1/2}$ strips away *all* the trivial kinematics (the phase-space integral f) and constants, leaving only the physics — the weak coupling G and the nuclear matrix element $|M_{fi}|^2$. Measuring $t_{1/2}$ and computing f thus delivers $|M_{fi}|^2$ directly: ft is the great equaliser that lets us compare decays across the chart.

The Weak Coupling Constant G

Each of the four forces has a “charge” setting how strongly matter responds to it: mass for gravity, electric charge Q for electromagnetism, and a coupling g for each of the strong and weak forces. In β decay G plays this role. It is fixed *empirically* from measured β lifetimes (via (208)); the result is ~ 2 orders of magnitude smaller than the strong coupling — which is precisely why we call it the *weak* force.

Using $\log ft$ in Practice

Because ft ranges over many orders of magnitude, one tabulates $\log_{10} ft$, typically between 3 and 20:

$$\log ft \approx 3 \text{ (very probable)} \quad \longrightarrow \quad \log ft \approx 20 \text{ (extremely improbable)}. \quad (209)$$

A small $\log ft$ means a large matrix element (strong overlap of initial and final nuclear wavefunctions); a large $\log ft$ flags a strongly suppressed (forbidden) transition.

11.6 Selection Rules: Fermi, Gamow–Teller, and Forbidden Transitions

Conceptual Background

Unlike the *absolute* selection rules of α and γ decay (where “forbidden” means truly forbidden), the β -decay rules are *quantitative*: they govern *how probable* a transition is,

not whether it can happen at all. Within the allowed approximation the leptons carry off no orbital angular momentum, $L = 0$, so the only angular momentum available to change the nuclear spin is the *coupled lepton spin* S of the $e\text{--}\bar{\nu}$ pair.

Physical Concept 11.4: Allowed Selection Rules

$$\textbf{Fermi: } S = 0 \text{ (} e^\uparrow \bar{\nu}^\downarrow \text{ or } e^\downarrow \bar{\nu}^\uparrow \text{), } \Delta I = 0, \quad \Delta\pi = \text{no.} \quad (210)$$

$$\textbf{Gamow\text{--}Teller: } S = 1 \text{ (parallel lepton spins), } \Delta I = 0, \pm 1, \quad \Delta\pi = \text{no.} \quad (211)$$

Significance: the lepton pair leaves either as a spin singlet (Fermi) or a spin triplet (Gamow–Teller). In *both* cases there is no parity change, because $\Delta\pi = (-1)^L$ and $L = 0$. (Gamow–Teller cannot drive a $0 \rightarrow 0$ transition because the triplet must change I by at least the unit it carries.)

Forbidden and Super-Allowed Transitions

Beyond the leading term, the Taylor expansion (196) supplies $L \neq 0$ contributions — *forbidden* transitions. They are not strictly forbidden, merely suppressed (larger $\log ft$); the parity rule becomes

$$\Delta\pi = (-1)^L. \quad (212)$$

The hierarchy: *super-allowed* ($\log ft \sim 3$, near-perfect overlap of initial/final wavefunctions) $\rightarrow$ *allowed* $\rightarrow$ *first forbidden* ($\Delta\pi = \text{yes}$) $\rightarrow$ *second forbidden*, etc., each step adding a unit of L and several orders of magnitude to ft . When all allowed channels are blocked by spin/parity, a forbidden transition becomes the *dominant* decay.

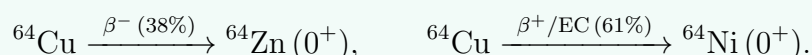
Why $L \neq 0$ is hard — a unified picture. The assumption $pR/\hbar \ll 1$ is *the same* as $L \approx 0$: classically $\mathbf{L} = \mathbf{r} \times \mathbf{p}$, so $|\mathbf{L}|/\hbar \lesssim pR/\hbar \ll 1$. To build even one quantum of orbital angular momentum the electron would have to be emitted far from the nuclear centre, but the nucleus is tiny compared with the lever arm needed for $L \sim \hbar$ at $p \sim 1 \text{ MeV}/c$. Quantum mechanically the lepton wavefunction extends to infinity, so $L > 0$ is not impossible — just exponentially unlikely, exactly as tunnelling makes high- ℓ α decay rare.

Method — Classifying a Given β Transition

1. Read off I_i^π and I_f^π of parent and daughter levels.
2. Compute $\Delta I = |I_i - I_f|$ and whether parity changes.
3. If $\Delta\pi = \text{no}$ and $\Delta I = 0, \pm 1$: *allowed* (Fermi if $\Delta I = 0$, Gamow–Teller if $\Delta I = 0, 1$).
4. If $\Delta\pi = \text{yes}$: at least *first forbidden* ($L = 1$).
5. Match the smallest L consistent with $\Delta\pi = (-1)^L$ and $|I_i - I_f| \leq L + S$; that order sets the expected $\log ft$.

Worked Example 11.1 (Worked Example: the ^{64}Cu Branches)

^{64}Cu ($I^\pi = 1^+$) sits at the floor of an $A = 64$ valley with stable neighbours on *both* sides, so it decays two ways:



Both are $1^+ \rightarrow 0^+$: $\Delta I = 1$, $\Delta\pi = \text{no} \Rightarrow$ *allowed Gamow–Teller* (forbidden as pure Fermi, which needs $\Delta I = 0$). The β^- spectrum follows the bare phase-space shape $N(p) \propto p^2(Q - T_e)^2$, while the β^+ spectrum is pushed to higher momenta by the Coulomb repulsion encoded in $F(Z', p) < 1$ at low p . *Energetics*: since EC appears, $Q_{\text{EC}} > 0$; β^+ runs only because $Q_{\text{EC}} > 1.022$ MeV, i.e. (192) is satisfied with margin.

11.7 Neutrino Physics from Beta Decay

(a) Neutrino Mass from the Spectrum Endpoint

We assumed $m_\nu = 0$ in (202). To test it, reinstate a small mass: it deforms the spectrum only at the extreme *high-energy endpoint*, where the electron takes nearly all the energy and the neutrino is left almost at rest. The shape there changes from meeting the axis tangentially ($m_\nu = 0$) to dropping more steeply ($m_\nu > 0$), but the effect is tiny — a shift comparable to the neutrino mass, tens of eV at most, against an electron mass of 511 keV.

$$m_\nu c^2 \lesssim 0.25 \text{ eV} \quad (\text{current bound}), \quad (213)$$

versus ~ 20 eV in older textbooks. No experiment has resolved a *nonzero* mass from the endpoint; one can only place ever-tighter upper bounds (e.g. tritium-endpoint and $0\nu\beta\beta$ searches). The Standard Model is built assuming $m_\nu = 0$, so a definite mass is one of the most important open questions in physics.

Note:

Pitfall: the endpoint distortion is a property of the *last few keV* of the spectrum only — the bulk shape is completely insensitive to m_ν . This is why these experiments demand enormous statistics and energy resolution.

(b) Helicity

Physical Concept 11.5: Helicity

For any spinning particle, the projection of its spin onto its momentum direction (up to normalisation):

$$h = \frac{\mathbf{S} \cdot \hat{\mathbf{p}}}{|\mathbf{S}|}. \quad (214)$$

$h > 0$ (spin along motion) is “right-handed”; $h < 0$ is “left-handed” (the right-hand/screw rule).

Naively one expects a 50/50 split, but experiment is startling:

$$\boxed{\text{all neutrinos are left-handed; all antineutrinos are right-handed.}} \quad (215)$$

Significance: the weak interaction *maximally violates parity*. There are no right-handed neutrinos to drive “inverse” weak reactions that would need them, a fact established beyond doubt and built into the Standard Model.

(c) The Solar Neutrino Problem → Neutrino Oscillations

The Sun shines by the pp -chain (four protons $\rightarrow$ ${}^4\text{He}$), emitting copious electron neutrinos ν_e . Davis's Homestake experiment (1.5 km underground, a huge ${}^{37}\text{Cl}$ tank) detected them via



counting ${}^{37}\text{Ar}$ atoms at a rate of ~ 1 atom/day over a decade. The result was a shock: only *one third* of the predicted flux. Two possibilities — either the solar model was wrong, or the neutrino theory was.

Physical Law 11.3 Resolution — Neutrino Oscillations

The Sun emits pure ν_e , but *only* ν_e can convert ${}^{37}\text{Cl}$. En route to Earth the state evolves into an equal mixture

$$|\nu_e\rangle \longrightarrow \frac{1}{3} |\nu_e\rangle + \frac{1}{3} |\nu_\mu\rangle + \frac{1}{3} |\nu_\tau\rangle,$$

so a ${}^{37}\text{Cl}$ detector sees only the surviving $\frac{1}{3}$.

Significance: flavour oscillation requires the flavour eigenstates to *not* be mass eigenstates — a mixing that is possible *only if neutrinos have mass*. Davis's deficit is thus direct evidence for $m_\nu \neq 0$, contradicting the massless-neutrino Standard Model. (Whether the neutrino is its own antiparticle — Dirac vs. Majorana — remains open, and is the target of $0\nu\beta\beta$ experiments.)

11.8 Gamma Decay Selection Rules (Brief)

Conceptual Background

Closing the radioactivity chapter, the lecturer states the γ (photon) multipole selection rules, contrasting them with β . A photon *always* carries away at least one unit of angular momentum:

$$|I_i - I_f| \leq L \leq I_i + I_f, \quad L \geq 1. \quad (217)$$

Hence a $0 \rightarrow 0$ electromagnetic transition is impossible — unlike β or α , where leptons/ α can leave radially with $L = 0$. A blocked $0 \rightarrow 0$ de-excitation must instead proceed by a *cascade* through intermediate levels or by *internal conversion* (handing the energy directly to an atomic electron).

Physical Concept 11.6: Electric vs. Magnetic Multipoles

Parity fixes the character (E = electric, M = magnetic) of each multipole order L :

$$\Delta\pi = (-1)^L : \quad \text{electric } EL, \quad (218)$$

$$\Delta\pi = (-1)^{L+1} : \quad \text{magnetic } ML. \quad (219)$$

Significance: for a given $\Delta\pi$ between two levels, E and M multipoles of the *same* order can never compete (parity forbids it); since each higher multipole is weaker by several orders of magnitude, the lowest allowed multipole dominates, giving a clean E/M hierarchy for every transition.

11.9 Putting It Together: ^{23}Al Decay Spectroscopy

Physical Motivation

The lecturer closes with a real study (I. Goldberg, Friedman group): β -decay spectroscopy of ^{23}Al ($t_{1/2} \approx 0.45$ s, $Q \approx 12$ MeV), illustrating how all the chapter's tools combine to map a nuclear level scheme.

Method — Reconstructing a Level Scheme from Decays

1. **Measure the half-life:** accumulate the activity for ~ 0.5 s, then watch the exponential decay — recovering $t_{1/2}$ and confirming build-up/decay kinetics from Lecture 5.
2. **Record the γ spectrum:** hundreds of lines on a log scale. Identify each line's origin, including detector artefacts:
 - a strong 511 keV line — from *both* pair production *and* positron annihilation following β^+ ;
 - *Single-Escape* (SE) peaks at $E_\gamma - 511$ keV and *Double-Escape* (DE) at $E_\gamma - 1022$ keV, when one or both annihilation photons leave the detector (e.g. a full-energy peak at 4082 keV shows an SE companion at 3571 keV).
3. **Subtract backgrounds:** identify ubiquitous room lines (K, Bi, U natural activity).
4. **Use coincidences:** γ -rays detected *simultaneously* belong to a *cascade* (emitted together down a chain). Proton coincidences flag β -delayed proton emission, possible when the daughter level lies above the proton separation energy.
5. **Assign I^π and build the scheme:** combine γ energies, coincidences, β branchings, and $\log ft$ to tag each level's spin and parity. A small $\log ft \approx 3.27$ flags an allowed/super-allowed β branch (very similar parent/daughter spin); larger values flag forbidden branches.

Significance: the full nuclear level structure — energies, spins, parities, decay paths (including β -delayed proton emission to ^{22}Na) — is reconstructed *purely* from its radioactive emissions. This is the practical payoff of the entire decay chapter: decays as a *probe* of nuclear structure.

Note:

Exam note: the TA homework is deliberately a review of these methods. The exam will be based *closely* on the homework problems — possibly with small variations — so memorising every derivation matters less than being fluent with the master formulas (spectrum shape, ft , selection-rule classification, Q -value energetics) and the worked methods above.

Homework 1: Nuclear Properties and Binding Energy

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Exercise Questions

1. Compute the form factors $F(q)$ for the following charge distribution:

$$\rho(r) = \rho_0 e^{-(\ln 2)r^2/R^2}.$$

Relevant formulas:

- Normalized form factor:

$$F(\mathbf{q}) = \frac{1}{Q} \int \rho(\mathbf{r}) e^{i\mathbf{q} \cdot \mathbf{r}} d^3r, \quad Q = \int \rho(\mathbf{r}) d^3r.$$

- For a spherically symmetric distribution:

$$F(q) = \frac{4\pi}{Q} \int_0^\infty r^2 \rho(r) \frac{\sin(qr)}{qr} dr.$$

- Gaussian transform identity, with $\alpha = \ln(2)/R^2$:

$$\int e^{-\alpha r^2} e^{i\mathbf{q} \cdot \mathbf{r}} d^3r = \left(\frac{\pi}{\alpha}\right)^{3/2} e^{-q^2/(4\alpha)}.$$

Solution:

Due to the spherical symmetry of the charge distribution, this integral simplifies to:

$$F(q) = 4\pi \int_0^\infty \frac{\sin(qr)}{qr} \rho(r) r^2 dr = \frac{4\pi}{q} \int_0^\infty \sin(qr) \rho(r) r dr. \quad (220)$$

Substituting the given charge density $\rho(r) = \rho_0 e^{-(\ln 2)r^2/R^2}$, We define the integral:

$$I = \int_0^\infty \sin(qr) e^{-(\ln 2)r^2/R^2} r dr. \quad (221)$$

This integral can be evaluated by rewriting the integrand as a derivative. Notice that $r \sin(qr) = -\frac{\partial}{\partial q} \cos(qr)$. Thus, we can rewrite the integral as:

$$I = -\frac{\partial}{\partial q} \int_0^\infty \cos(qr) e^{-\alpha r^2} dr \quad (222)$$

Since the integrand is even, we can extend the integration limits to infinity:

$$\int_0^\infty \cos(qr) e^{-\alpha r^2} dr = \frac{1}{2} \int_{-\infty}^\infty \cos(qr) e^{-\alpha r^2} dr \quad (223)$$

Using Euler's formula, $\cos(qr)$ is the real part of e^{iqr} . Since the imaginary part integrates to zero (as it is an odd function), we have:

$$\frac{1}{2} \int_{-\infty}^\infty \cos(qr) e^{-\alpha r^2} dr = \frac{1}{2} \int_{-\infty}^\infty e^{iqr} e^{-\alpha r^2} dr \quad (224)$$

This is a standard 1D Gaussian integral, which evaluates to:

$$\frac{1}{2} \int_{-\infty}^\infty e^{iqr} e^{-\alpha r^2} dr = \frac{1}{2} \sqrt{\frac{\pi}{\alpha}} e^{-q^2/(4\alpha)} \quad (225)$$

Now, we differentiate this result with respect to q to find I :

$$I = -\frac{\partial}{\partial q} \left(\frac{1}{2} \sqrt{\frac{\pi}{\alpha}} e^{-q^2/(4\alpha)} \right) = -\frac{1}{2} \sqrt{\frac{\pi}{\alpha}} \left(-\frac{2q}{4\alpha} \right) e^{-q^2/(4\alpha)} \quad (226)$$

$$I = \frac{q}{4\alpha} \sqrt{\frac{\pi}{\alpha}} e^{-q^2/(4\alpha)}, \quad (227)$$

Where $\alpha = \frac{\ln 2}{R^2}$. Substituting back into the form factor expression:

$$F(q) = \frac{4\pi}{q} \cdot \rho_0 \cdot \left(\frac{q}{4\alpha} \sqrt{\frac{\pi}{\alpha}} e^{-q^2/(4\alpha)} \right). \quad (228)$$

Simplifying and substituting the value of α , we find:

$$F(q) = \rho_0 R^3 \left(\frac{\pi}{\ln 2} \right)^{3/2} e^{-\frac{q^2 R^2}{4 \ln 2}}. \quad (229)$$

A nuclear charge distribution more realistic than the uniformly charged distribution is the Fermi distribution,

$$\rho(r) = \rho_0 \{1 + \exp[(r - R)/a]\}^{-1}.$$

- (a) Sketch this distribution.
- (b) Find the value of a if $t = 2.3$ fm.
- (c) What is the significance of the parameter R ?
- (d) Evaluate $\langle r^2 \rangle$ according to this distribution.

Relevant formulas:

- Half-density radius:

$$\rho(R) = \frac{\rho_0}{2}.$$

- Surface thickness between $0.9\rho_0$ and $0.1\rho_0$:

$$t = r_{0.1} - r_{0.9} = 4a \ln 3.$$

- Mean-square radius:

$$\langle r^2 \rangle = \frac{\int r^2 \rho(\mathbf{r}) d^3r}{\int \rho(\mathbf{r}) d^3r} = \frac{\int_0^\infty r^4 \rho(r) dr}{\int_0^\infty r^2 \rho(r) dr}.$$

- Standard leading approximation for $a/R \ll 1$:

$$\langle r^2 \rangle \simeq \frac{3}{5}R^2 + \frac{7}{5}\pi^2 a^2.$$

- (a) From the known masses of ^{15}O and ^{15}N , compute the difference in binding energy.
- (b) Assuming this difference to arise from the difference in Coulomb energy, compute the nuclear radius of ^{15}O and ^{15}N .

Relevant formulas:

- Binding energy from atomic masses:

$$B(Z, N) = [Zm(^1\text{H}) + Nm_n - M_{\text{atom}}(Z, N)] c^2.$$

- Binding-energy difference:

$$\Delta B = B(^{15}\text{O}) - B(^{15}\text{N}).$$

- Coulomb energy of a uniformly charged sphere:

$$E_C(Z, R) = \frac{3}{5} \frac{Z(Z-1)e^2}{4\pi\epsilon_0 R} \simeq \frac{3}{5} \frac{Z(Z-1)(1.44 \text{ MeV fm})}{R}.$$

- Mirror-nucleus estimate:

$$\Delta E_C \simeq \frac{3}{5} \frac{1.44 \text{ MeV fm}}{R} [Z_{^{15}\text{O}}(Z_{^{15}\text{O}} - 1) - Z_{^{15}\text{N}}(Z_{^{15}\text{N}} - 1)].$$

- (c) Given the following mass doublet values (in units of 10^{-6}u), compute the corresponding values for the atomic mass of ^{37}Cl :

$$\begin{aligned} m(\text{C}_3\text{H}) - m(^{37}\text{Cl}) &= 41922.2 \pm 0.3, \\ m(\text{C}_2\text{D}_3) - m(^{37}\text{ClH}_3) &= 123436.5 \pm 0.1, \\ m(\text{C}_3\text{H}_6\text{O}_2) - m(^{37}\text{Cl}_2) &= 104974.24 \pm 0.08. \end{aligned}$$

Here $\text{D} \equiv {}^2\text{H}$, $\text{C} \equiv {}^{12}\text{C}$, and $\text{O} \equiv {}^{16}\text{O}$. Include in your calculation the effect of uncertainties in the H, D, O, and C masses.

Relevant formulas:

- Molecular-mass bookkeeping:

$$m(\text{C}_3\text{H}) = 3m(\text{C}) + m(\text{H}), \quad m(^{37}\text{ClH}_3) = m(^{37}\text{Cl}) + 3m(\text{H}).$$

- Solve each doublet equation algebraically for $m(^{37}\text{Cl})$. For example,

$$m(^{37}\text{Cl}) = m(\text{C}_3\text{H}) - (41922.2 \times 10^{-6}) \text{ u}.$$

- The three corresponding mass estimates are

$$\begin{aligned} m_1(^{37}\text{Cl}) &= 3m(\text{C}) + m(\text{H}) - \Delta_1, \\ m_2(^{37}\text{Cl}) &= 2m(\text{C}) + 3m(\text{D}) - 3m(\text{H}) - \Delta_2, \\ m_3(^{37}\text{Cl}) &= \frac{1}{2} [3m(\text{C}) + 6m(\text{H}) + 2m(\text{O}) - \Delta_3], \end{aligned}$$

where each Δ_i is the quoted doublet value multiplied by 10^{-6} u.

- Independent uncertainty propagation:

$$\sigma_f^2 = \sum_i \left(\frac{\partial f}{\partial x_i} \sigma_{x_i} \right)^2.$$

- Weighted average of independent mass estimates:

$$\bar{m} = \frac{\sum_i m_i / \sigma_i^2}{\sum_i 1 / \sigma_i^2}, \quad \sigma_{\bar{m}} = \left(\sum_i \frac{1}{\sigma_i^2} \right)^{-1/2}.$$

- (d) Use the semiempirical mass formula to obtain an expression for the two-neutron separation energy when $A \gg 1$. (Hint: A differential method is far easier than an algebraic one for this problem.) Estimate the size of the various terms and discuss the A dependence. Compare with the following data for Al and Te:

^{25}Al	31.82 MeV	^{117}Te	18.89 MeV
^{26}Al	28.30 MeV	^{118}Te	18.45 MeV
^{27}Al	24.42 MeV	^{119}Te	18.17 MeV
^{28}Al	20.78 MeV	^{120}Te	17.88 MeV
^{29}Al	17.16 MeV	^{121}Te	17.46 MeV
^{30}Al	15.19 MeV	^{122}Te	17.04 MeV
^{31}Al	13.03 MeV	^{123}Te	16.80 MeV

Why do we choose two-neutron, rather than one-neutron, separation energies for this comparison?

Relevant formulas:

- Semiempirical mass formula:

$$B(A, Z) = a_v A - a_s A^{2/3} - a_c \frac{Z(Z-1)}{A^{1/3}} - a_a \frac{(A-2Z)^2}{A} + \delta(A, Z).$$

- Two-neutron separation energy:

$$S_{2n}(A, Z) = B(A, Z) - B(A-2, Z).$$

- Differential approximation for $A \gg 1$:

$$S_{2n}(A, Z) \simeq 2 \left(\frac{\partial B}{\partial A} \right)_Z.$$

- Useful derivative at fixed Z , neglecting the derivative of the pairing term:

$$\left(\frac{\partial B}{\partial A} \right)_Z = a_v - \frac{2}{3}a_s A^{-1/3} + \frac{1}{3}a_c Z(Z-1)A^{-4/3} - a_a \left[1 - \frac{4Z^2}{A^2} \right].$$

- Pairing term scale:

$$\delta(A, Z) \sim \begin{cases} +a_p A^{-1/2}, & \text{even-even,} \\ 0, & \text{odd } A, \\ -a_p A^{-1/2}, & \text{odd-odd.} \end{cases}$$

- (e) The spin-parity of ${}^9\text{Be}$ and ${}^9\text{B}$ are both $\frac{3}{2}^-$. Assuming in both cases that the spin and parity are characteristic only of the odd nucleon, show how it is possible to obtain the observed spin-parity of ${}^{10}\text{B}$ (3^+). What other spin-parity combinations could also appear? Compare your results to the first few excited states of ${}^{10}\text{B}$.

Relevant formulas:

- Angular-momentum coupling:

$$J = |j_1 - j_2|, |j_1 - j_2| + 1, \dots, j_1 + j_2.$$

- Parity multiplication:

$$\pi = \pi_1 \pi_2 = (-1)^{\ell_1 + \ell_2}.$$

- For two odd nucleons with $j_1 = j_2 = \frac{3}{2}$ and $\pi_1 = \pi_2 = -$:

$$J = 0, 1, 2, 3, \quad \pi = (-)(-) = +.$$

- Allowed spin-parity multiplet:

$$J^\pi = 0^+, 1^+, 2^+, 3^+.$$